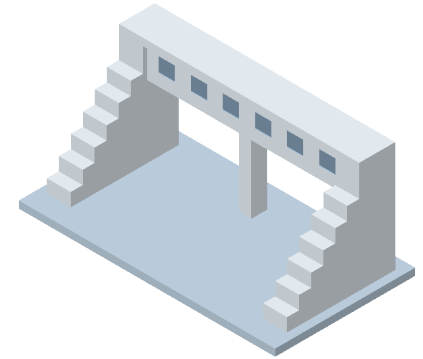
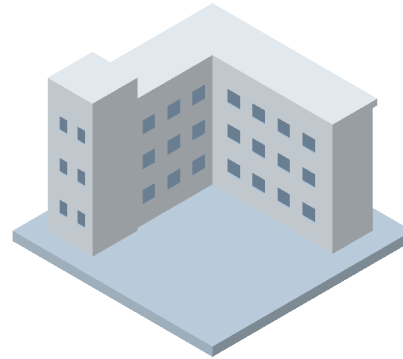
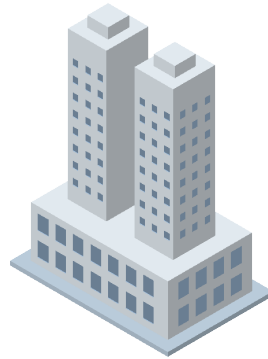


Turn Hong Kong buildings into your 3D community

Form 2 Tinkercad | 16 building references with three-view dimensions

Student pack



Choose one building, study its form, three views and arrow dimensions, then build it in Tinkercad.

01 Choose a model

Choose a residential, commercial, school or footbridge reference on the next pages.

02 Build from dimensions

All dimensions use mm. Make the main form first, then add windows and other details.

03 Finish and submit

Check that every part is connected, then submit the STL and a screenshot.

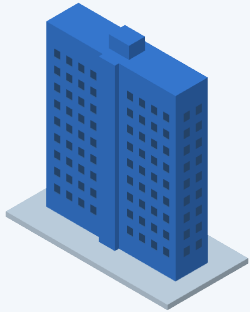
Class: _____ Name: _____ Group: _____

Choose a model: housing and mixed use

Model menu

1 Starter | 2 Developing | 3 Challenge. First build: R1 or S1.

R1 Straight corridor housing



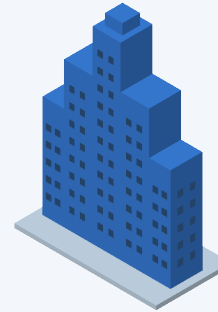
Level 1 • Page 4

R2 Cross-shaped public housing



Level 2 • Page 7

R3 Stepped residential building



Level 2 • Page 10

R4 H-shaped housing



Level 2 • Page 13

R5 Twin-tower housing



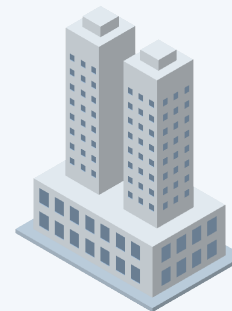
Level 3 • Page 16

M1 Long residential block + street shops



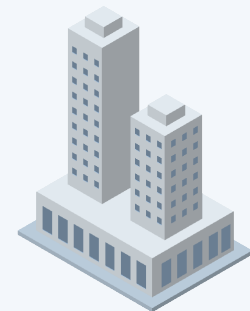
Level 2 • Page 19

M2 Twin residential towers + mall



Level 2 • Page 22

M3 High/low housing + corner shops



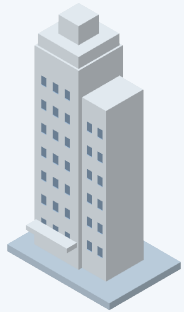
Level 2 • Page 25

Choose a model: commercial, schools and footbridges

1 Starter | 2 Developing | 3 Challenge. First build: R1 or S1.

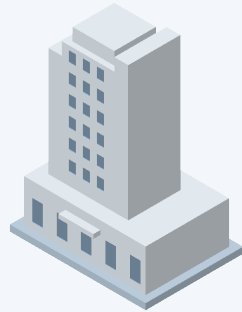
Model menu

C1 Modern office tower



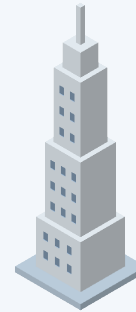
Level 2 • Page 28

C2 Mall-style commercial building



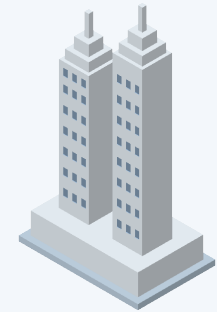
Level 2 • Page 31

C3 Slender pointed skyscraper



Level 3 • Page 34

C4 Twin glass towers



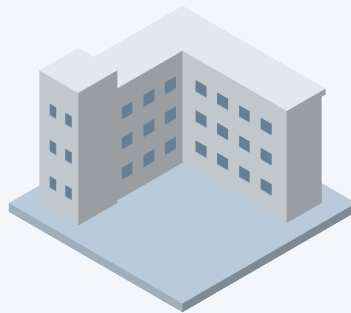
Level 3 • Page 37

S1 Secondary school main block



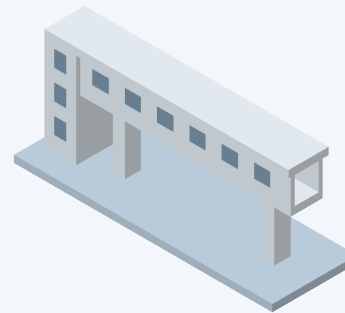
Level 1 • Page 40

S2 L-shaped school campus



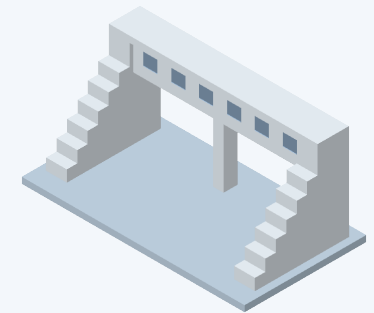
Level 2 • Page 43

B1 Straight covered footbridge



Level 2 • Page 46

B2 Covered footbridge with stairs



Level 3 • Page 49

R1 Straight corridor housing

Use the reference shape; dimensions are Width × Depth × Height (mm).

Residential

香港住宅大廈 (一字型長廊樓)

建築外觀示意

Tinkercad 建模示意



AI 生成概念圖

Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | A long main block with a central strip and repeated windows.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base plate	80 × 40 × 2	×1
Long main block	64 × 16 × 66	×1
Central strip	8 × 2 × 66	×1
Roof plant room	10 × 8 × 5	×1
Window-hole module (reused)	3 × 1 × 3	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

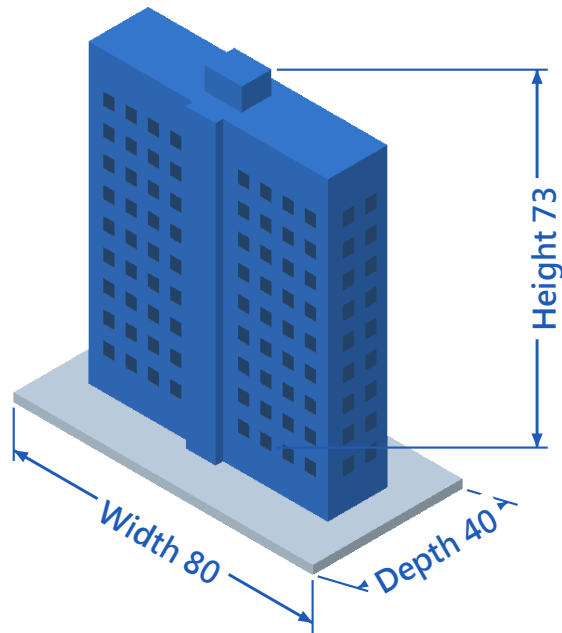
Print note | Check walls, bridges and roof parts with the teacher before printing.

R1 Three views and dimensions

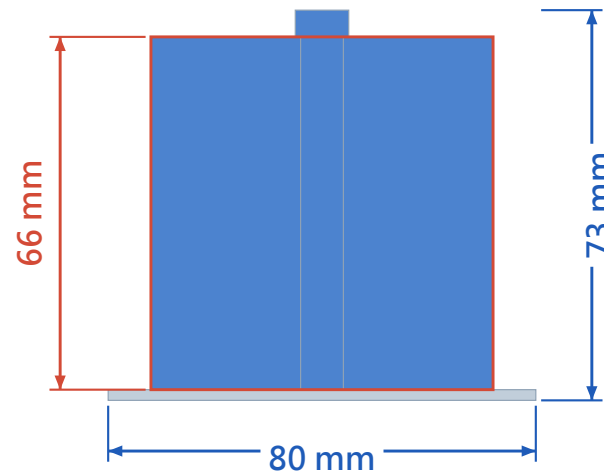
Dimension reference

Blue: overall (including base) | Red: Long main block | Unit: mm; see the previous page for other parts.

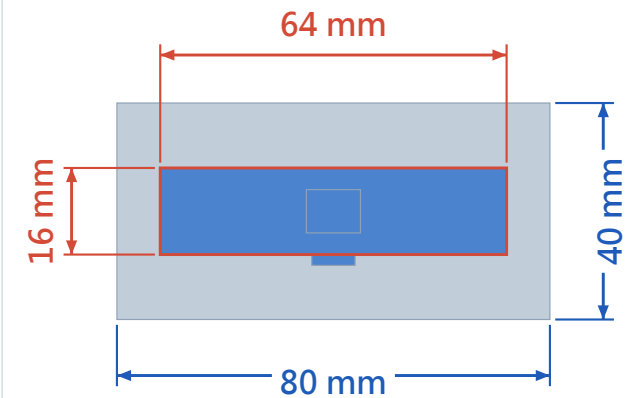
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = $80 \times 40 \times 73$ mm

Long main block

Width × Depth × Height = $64 \times 16 \times 66$ mm

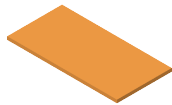
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

R1 Build it: Straight corridor housing

Residential

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

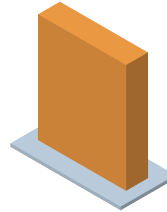
01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($80 \times 40 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

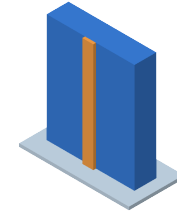
02 Place the long main block



Add the next Box using its listed size ($64 \times 16 \times 66$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

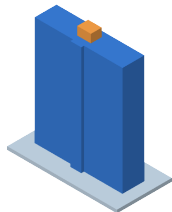
03 Add the central strip



Add or duplicate the next part ($8 \times 2 \times 66$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

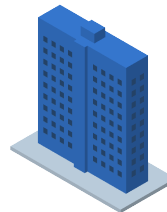
04 Add the roof plant room



Add the roof, connector or entrance part ($10 \times 8 \times 5$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

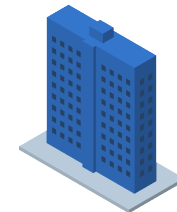
05 Arrange shallow window recesses



Make four window columns on the front and repeat them for nine rows. Add two columns on the side if time allows; keep the recesses shallow.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

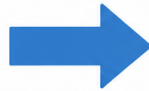
R2 Cross-shaped public housing

Use the reference shape; dimensions are Width × Depth × Height (mm).

Residential

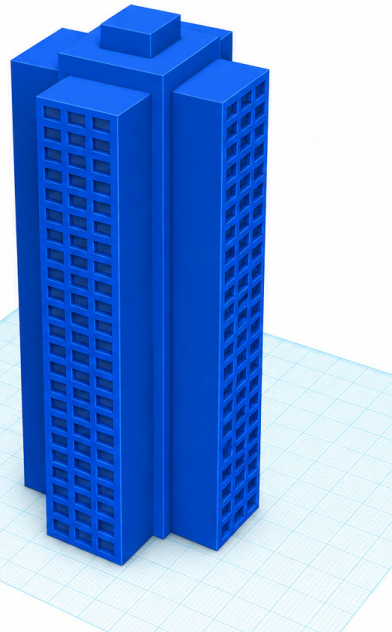
香港公屋住宅

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Build dimensions

Base plate	60 × 60 × 2	×1
Central tower	20 × 20 × 88	×1
Wings (front/rear turned 90°)	12 × 16 × 82	×4
Roof plant room	10 × 10 × 5	×1
Window-hole module (reused)	3 × 1 × 3	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

Print note | Check walls, bridges and roof parts with the teacher before printing.

Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | A central tower with four wings forming a cross in plan.

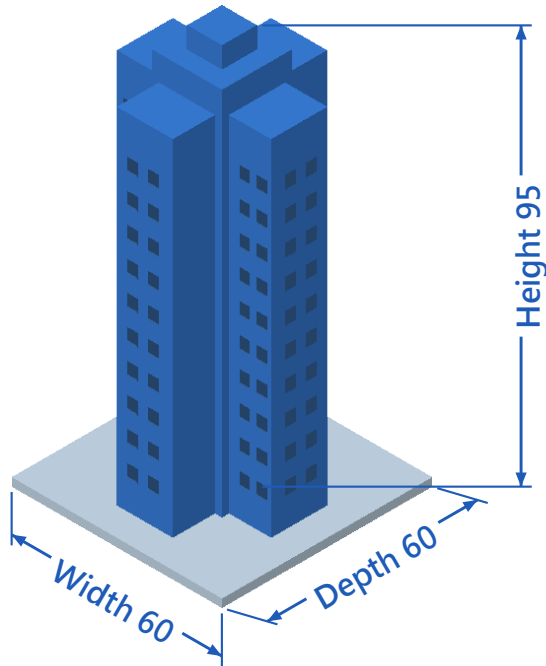
Basic version: complete the main form first. Add a few windows, then add more if time allows.

R2 Three views and dimensions

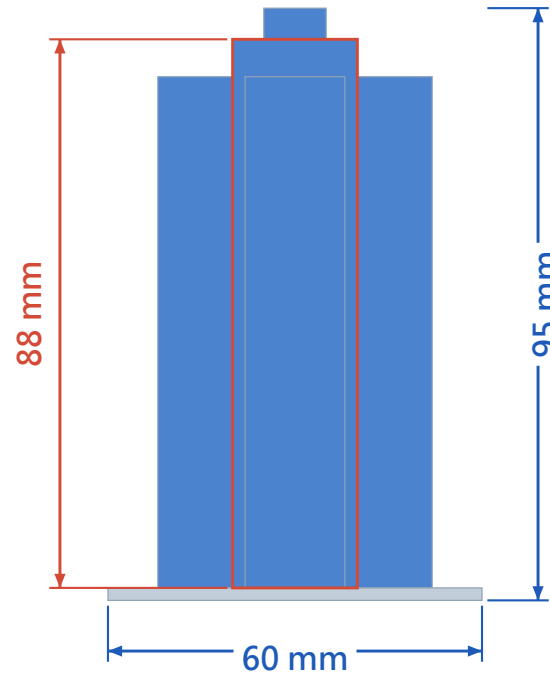
Dimension reference

Blue: overall (including base) | Red: Central tower | Unit: mm; see the previous page for other parts.

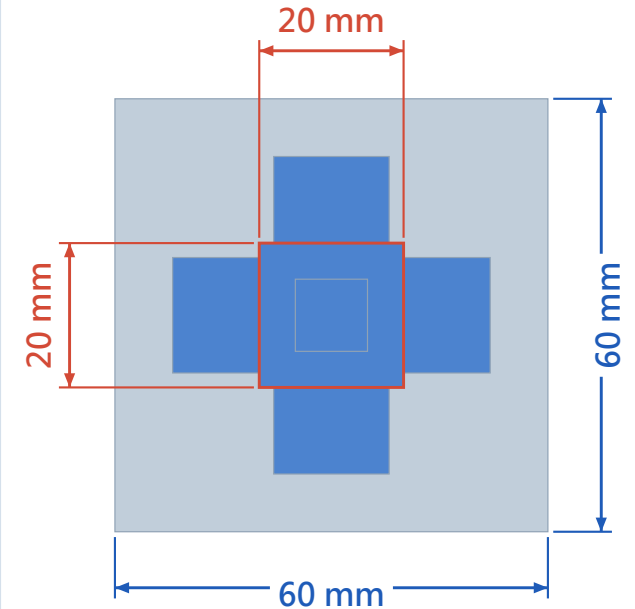
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 60 × 60 × 95 mm

Central tower

Width × Depth × Height = 20 × 20 × 88 mm

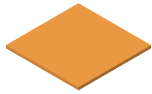
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

R2 Build it: Cross-shaped public housing

Residential

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

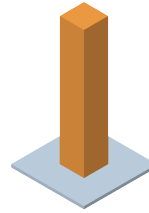
01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($60 \times 60 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

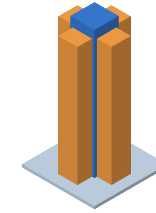
02 Make the central tower



Add the next Box using its listed size ($20 \times 20 \times 88$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

03 Add four wings



Add or duplicate the next part ($12 \times 16 \times 82$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

04 Add the roof plant room



Add the roof, connector or entrance part ($10 \times 10 \times 5$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

05 Make repeated windows



Make two window columns on each wing and repeat them for ten rows. Finish the visible front faces first.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

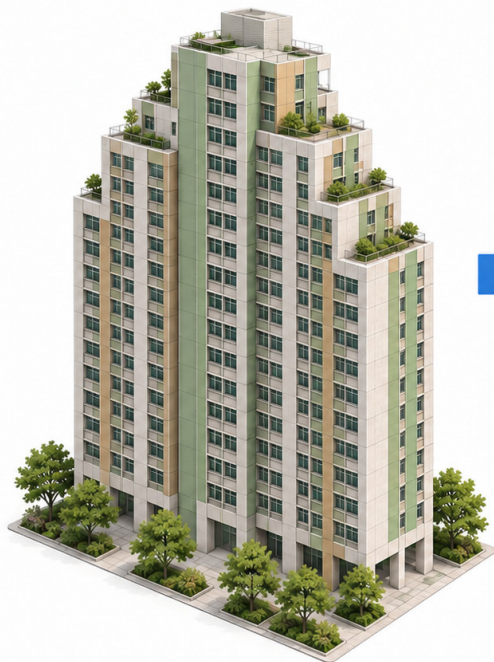
R3 Stepped residential building

Use the reference shape; dimensions are Width × Depth × Height (mm).

Residential

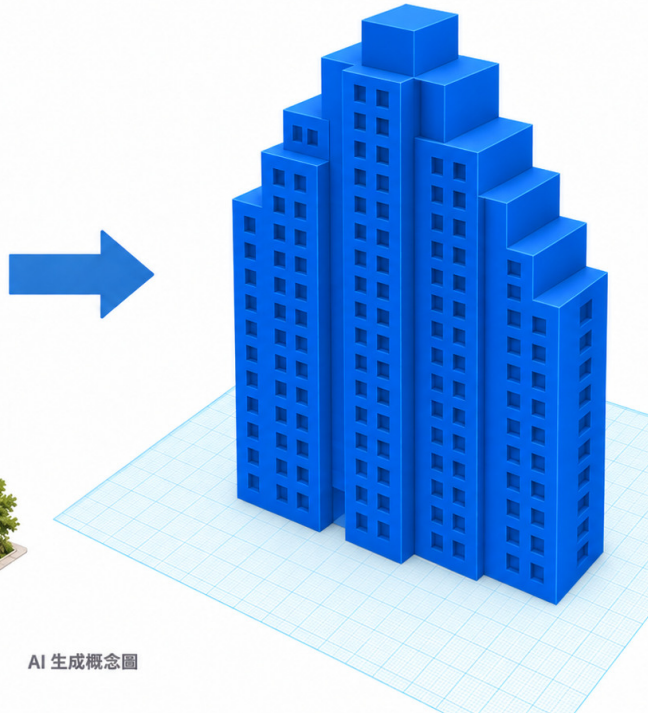
香港住宅大廈 (階梯式住宅)

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | The centre is highest and the two sides step down symmetrically.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base plate	80 × 42 × 2	×1
Central high block	16 × 18 × 88	×1
Middle blocks	16 × 18 × 68	×2
Outer low blocks	12 × 18 × 48	×2
Roof plant room	10 × 10 × 5	×1
Window-hole module (reused)	3 × 1 × 3	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

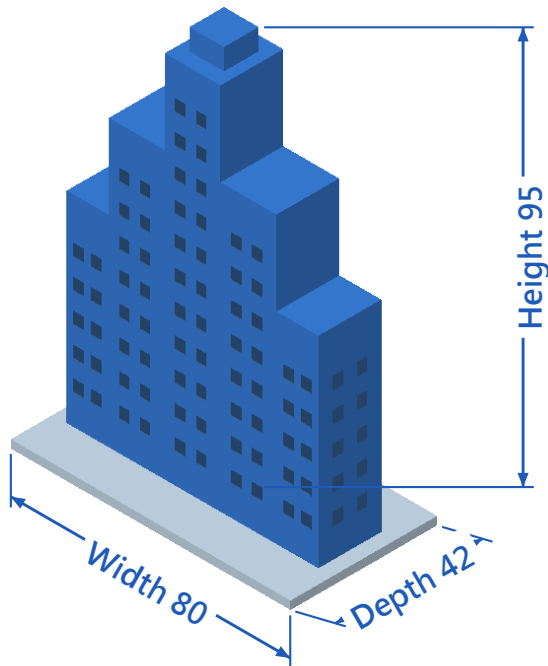
Print note | Check walls, bridges and roof parts with the teacher before printing.

R3 Three views and dimensions

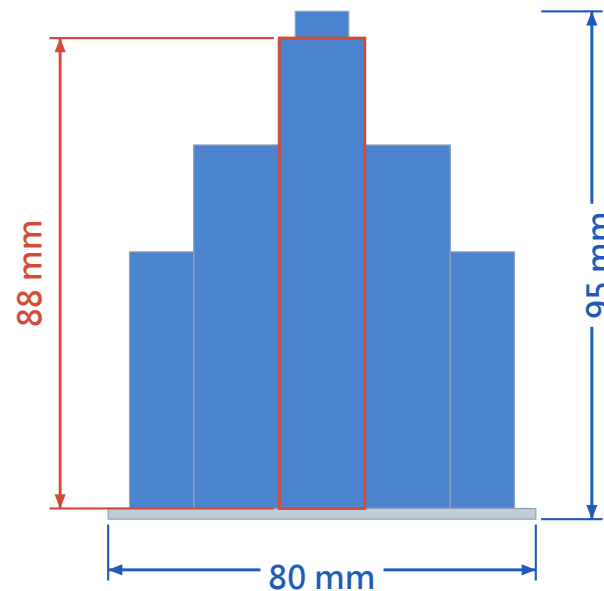
Dimension reference

Blue: overall (including base) | Red: Central high block | Unit: mm; see the previous page for other parts.

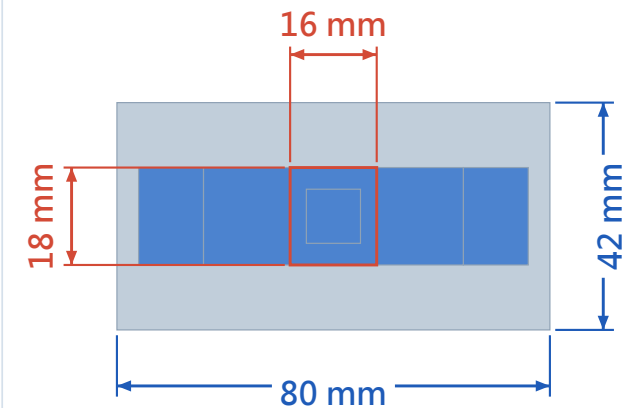
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 80 × 42 × 95 mm

Central high block

Width × Depth × Height = 16 × 18 × 88 mm

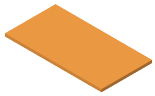
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

R3 Build it: Stepped residential building

Residential

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

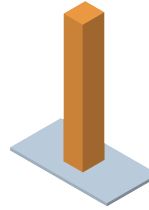
01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($80 \times 42 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

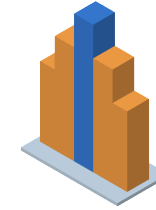
02 Place the highest block



Add the next Box using its listed size ($16 \times 18 \times 88$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

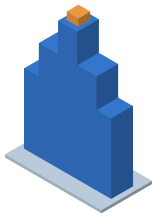
03 Arrange the stepped outline



Add or duplicate the next part ($16 \times 18 \times 68$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

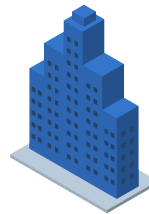
04 Add the top plant room



Add the roof, connector or entrance part ($12 \times 18 \times 48$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

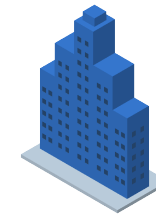
05 Make windows to suit each height



Make two front window columns. Use 11 rows on the high block, 8 on the middle blocks and 5 on the low blocks.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

R4 H-shaped housing

Use the reference shape; dimensions are Width × Depth × Height (mm).

Residential

香港住宅大廈 (H型住宅)



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | Two parallel wings are joined by a lower centre block.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base plate	80 × 60 × 2	×1
Side wings	18 × 40 × 80	×2
Central connector	28 × 14 × 56	×1
Roof plant room	8 × 8 × 5	×2
Window-hole module (reused)	3 × 1 × 3	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

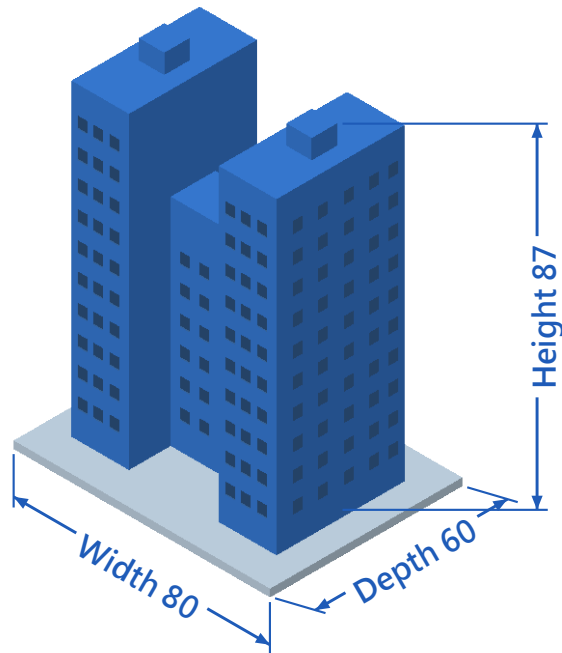
Print note | Check walls, bridges and roof parts with the teacher before printing.

R4 Three views and dimensions

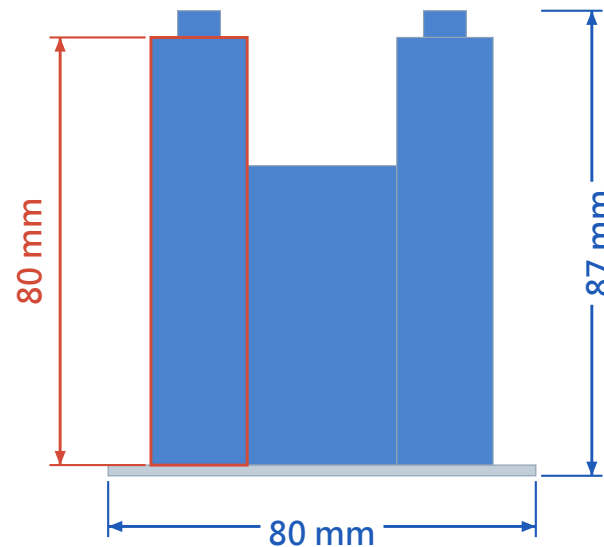
Dimension reference

Blue: overall (including base) | Red: Left wing | Unit: mm; see the previous page for other parts.

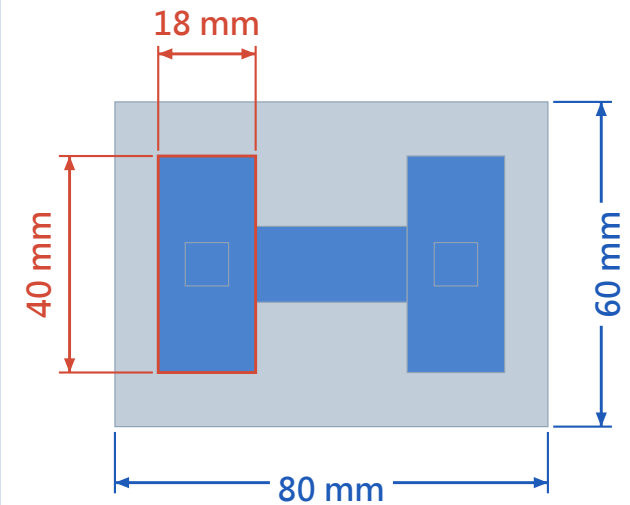
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 80 × 60 × 87 mm

Left wing

Width × Depth × Height = 18 × 40 × 80 mm

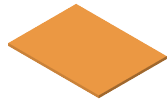
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

R4 Build it: H-shaped housing

Residential

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($80 \times 60 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

02 Place the two wings



Add the next Box using its listed size ($18 \times 40 \times 80$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

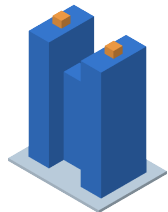
03 Connect the H shape



Add or duplicate the next part ($28 \times 14 \times 56$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

04 Add two roof plant rooms



Add the roof, connector or entrance part ($8 \times 8 \times 5$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

05 Add recessed windows



Make three columns by ten rows on each wing and four columns by six rows on the central connector.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

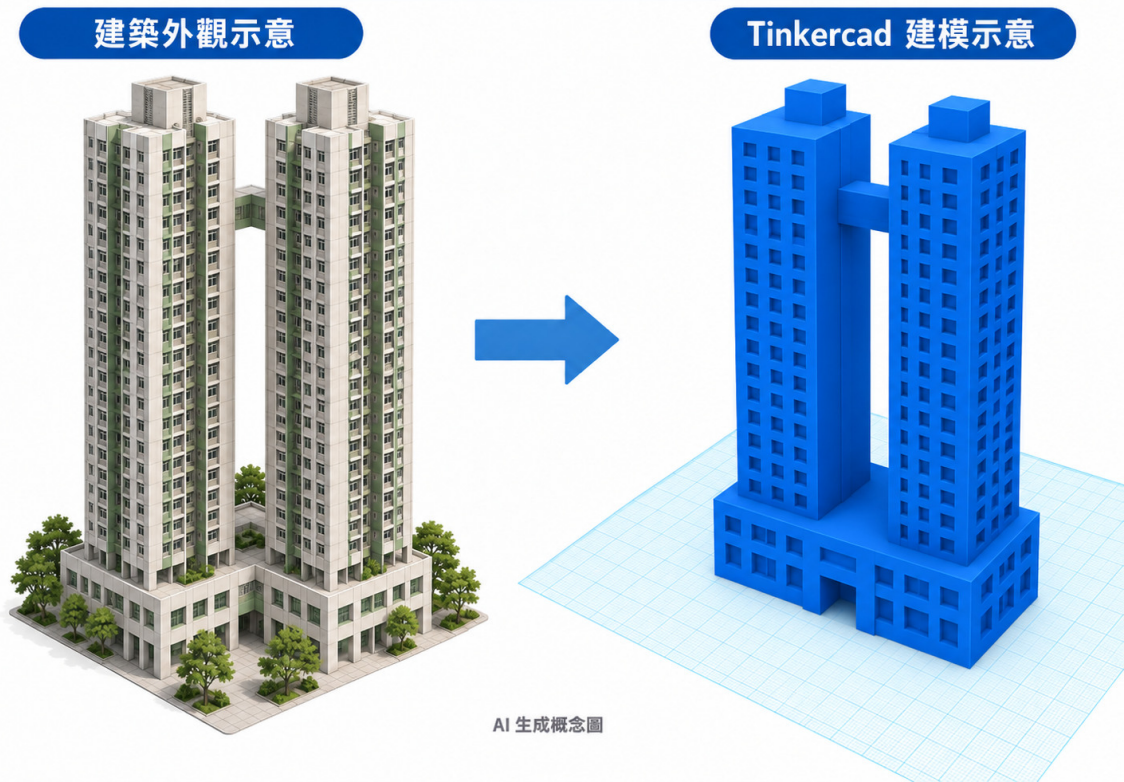
Check: The building is connected, with no floating pieces or unselected parts.

R5 Twin-tower housing

Use the reference shape; dimensions are Width × Depth × Height (mm).

Residential

香港住宅大廈（雙塔式住宅）



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | Two homes share a platform and a high connecting bridge.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base plate	80 × 48 × 2	×1
Platform	66 × 32 × 12	×1
Residential tower	18 × 22 × 70	×2
High bridge	20 × 10 × 5	×1
Roof plant room	8 × 10 × 5	×2
Window-hole module (reused)	3 × 1 × 3	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

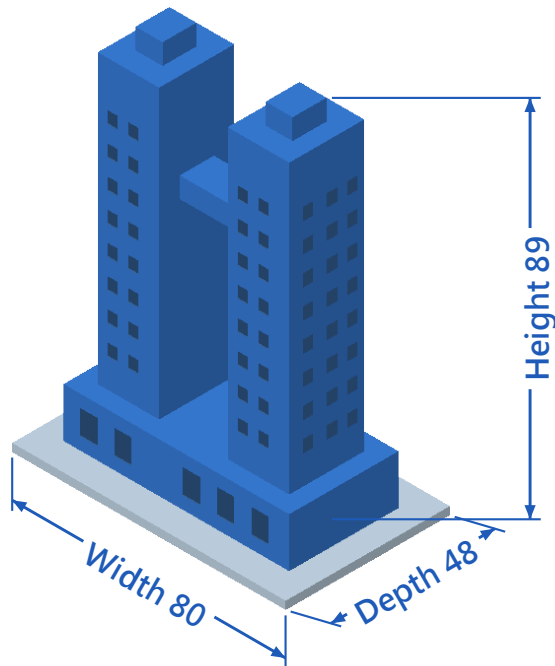
Print note | Check walls, bridges and roof parts with the teacher before printing.

R5 Three views and dimensions

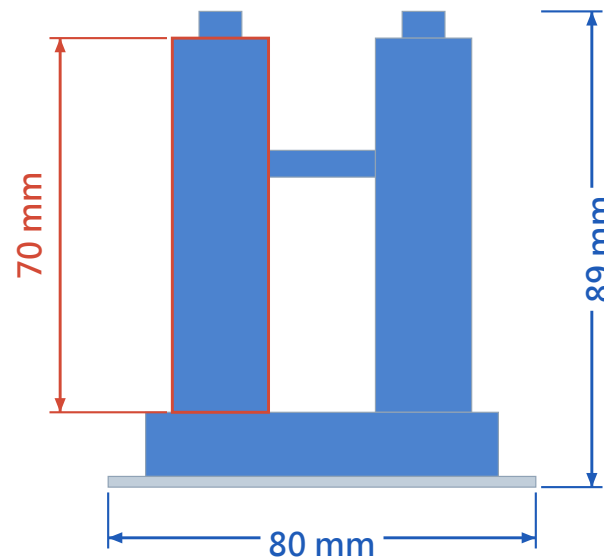
Dimension reference

Blue: overall (including base) | Red: Left tower | Unit: mm; see the previous page for other parts.

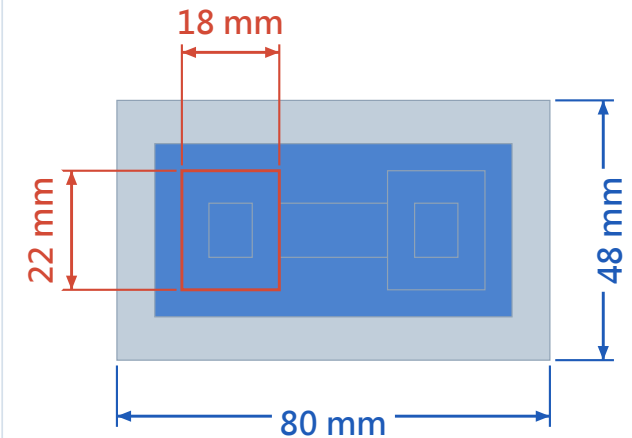
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 80 × 48 × 89 mm

Left tower

Width × Depth × Height = 18 × 22 × 70 mm

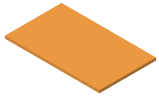
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

R5 Build it: Twin-tower housing

Residential

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

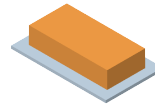
01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($80 \times 48 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

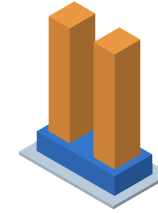
02 Make the shared platform



Add the next Box using its listed size ($66 \times 32 \times 12$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

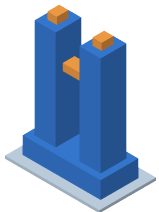
03 Place the two residential towers



Add or duplicate the next part ($18 \times 22 \times 70$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

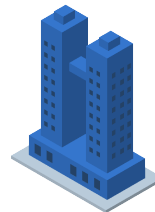
04 Add the bridge and plant rooms



Add the roof, connector or entrance part ($20 \times 10 \times 5$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

05 Add recessed windows



Make two columns by eight rows on each tower. Add a larger shallow entrance recess on the platform if time allows.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

M1 Long residential block + street shops

Mixed use

Use the reference shape; dimensions are Width × Depth × Height (mm).

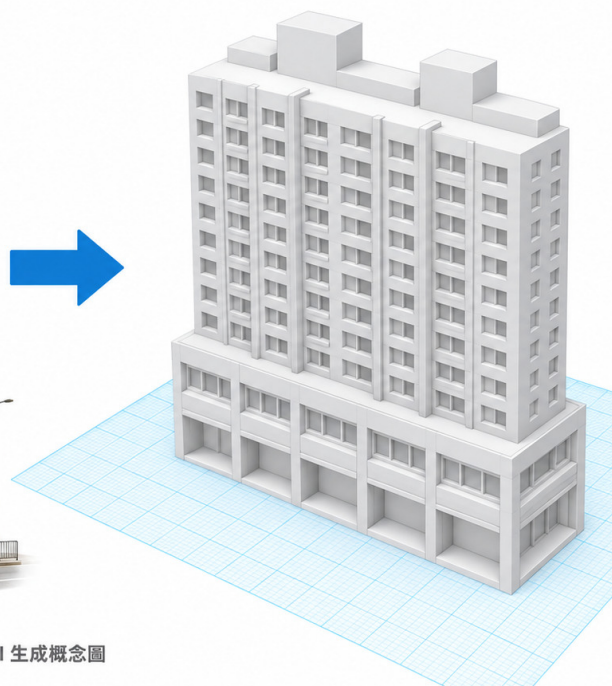
香港混合建築 (住宅 + 商舖)

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | A wider shop base supports one long residential block.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base plate	90 × 50 × 2	×1
Shop base	78 × 30 × 18	×1
Upper residential block	70 × 20 × 58	×1
Roof plant room	10 × 8 × 5	×2
Window-hole module (reused)	3 × 1 × 3	×1
Shopfront-hole module (reused)	6 × 1 × 11	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

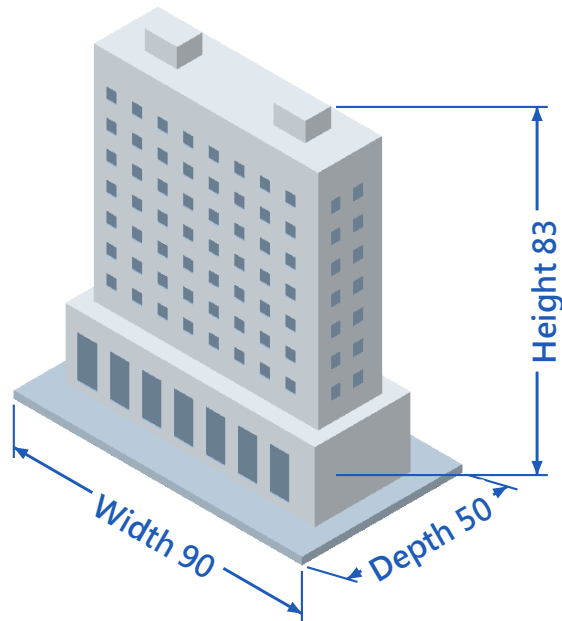
Print note | Check walls, bridges and roof parts with the teacher before printing.

M1 Three views and dimensions

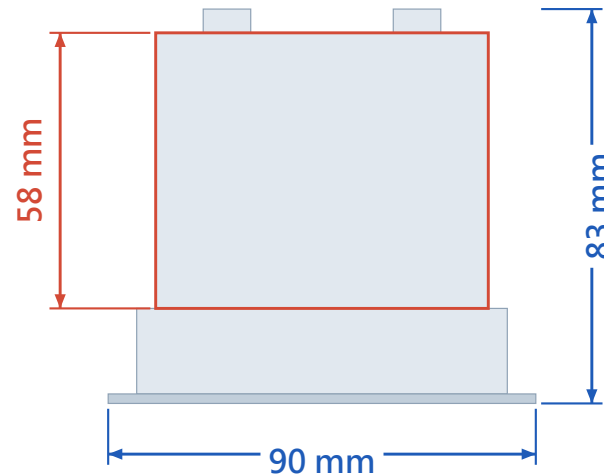
Dimension reference

Blue: overall (including base) | Red: Upper residential block | Unit: mm; see the previous page for other parts.

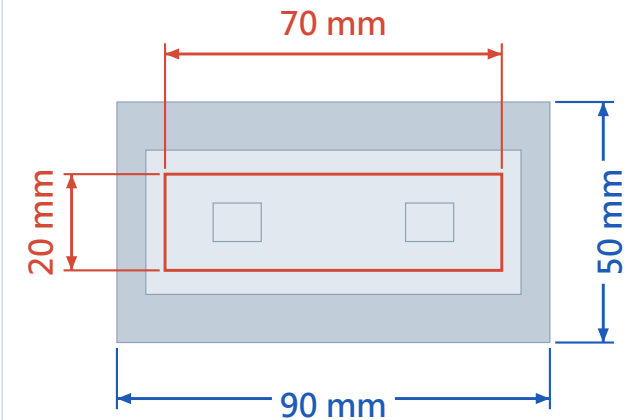
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 90 × 50 × 83 mm

Upper residential block

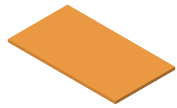
Width × Depth × Height = 70 × 20 × 58 mm

Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

M1 Build it: Long residential block + street shops Mixed use

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

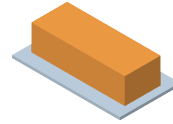
01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($90 \times 50 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

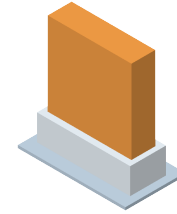
02 Place the shop base



Add the next Box using its listed size ($78 \times 30 \times 18$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

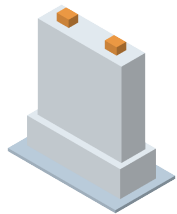
03 Place housing above the shops



Add or duplicate the next part ($70 \times 20 \times 58$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

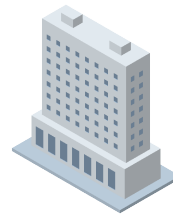
04 Add the roof plant room



Add the roof, connector or entrance part ($10 \times 8 \times 5$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

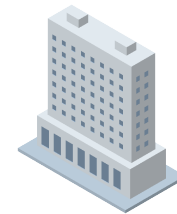
05 Separate windows and shopfronts



Make eight columns by seven rows of small residential windows. Use the $6 \times 1 \times 11$ mm tool for seven larger shopfront openings.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

M2 Twin residential towers + mall

Use the reference shape; dimensions are Width × Depth × Height (mm).

Mixed use

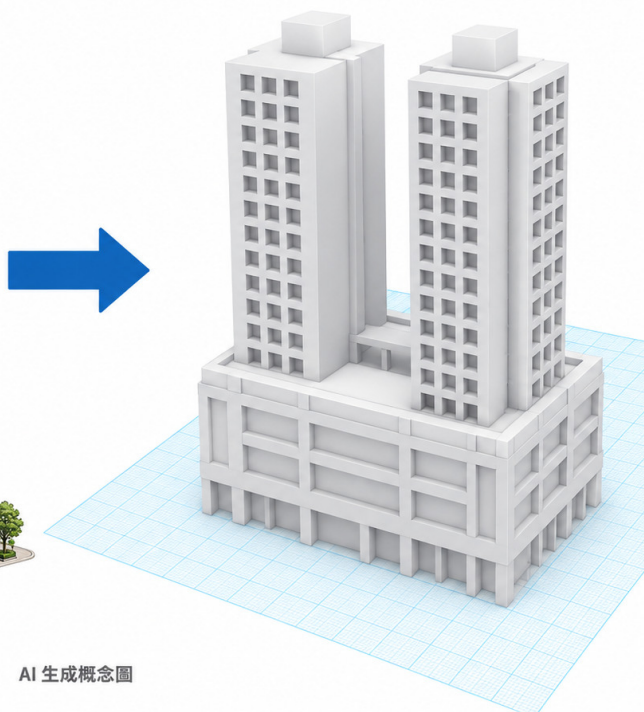
香港混合建築 (住宅 + 商舖)

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | A broad mall base supports two residential towers.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base plate	90 × 60 × 2	×1
Mall base	78 × 48 × 24	×1
Residential tower	22 × 24 × 70	×2
Roof plant room	12 × 12 × 5	×2
Window-hole module (reused)	3 × 1 × 3	×1
Mall window-hole module	6 × 1 × 7	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

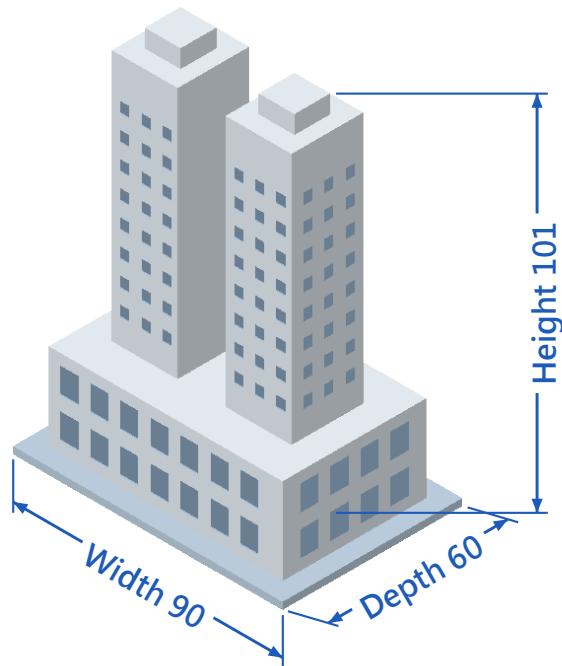
Print note | Check walls, bridges and roof parts with the teacher before printing.

M2 Three views and dimensions

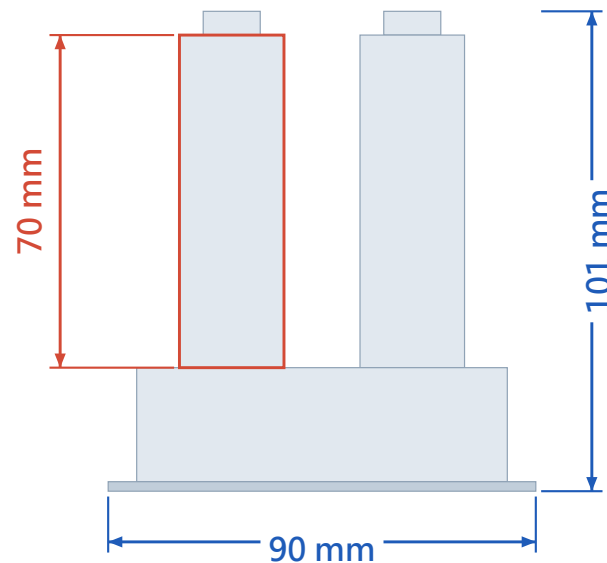
Dimension reference

Blue: overall (including base) | Red: Main part | Unit: mm; see the previous page for other parts.

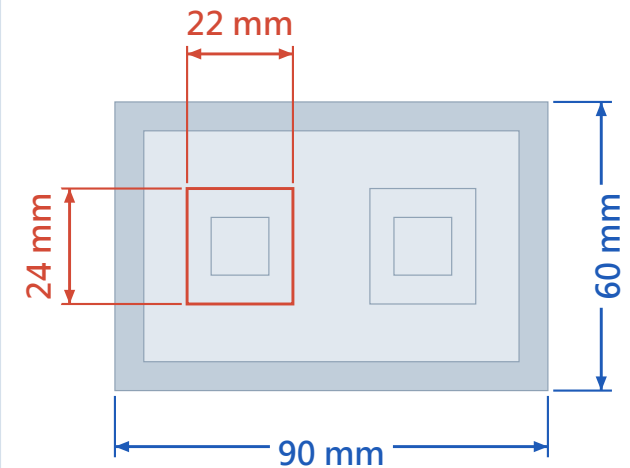
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 90 × 60 × 101 mm

Main part

Width × Depth × Height = 22 × 24 × 70 mm

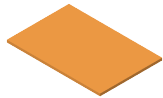
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

M2 Build it: Twin residential towers + mall

Mixed use

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

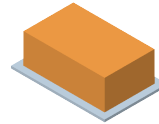
01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($90 \times 60 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

02 Build the large mall base



Add the next Box using its listed size ($78 \times 48 \times 24$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

03 Place two residential towers above it



Add or duplicate the next part ($22 \times 24 \times 70$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

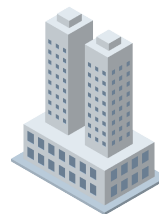
04 Add two roof plant rooms



Add the roof, connector or entrance part ($12 \times 12 \times 5$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

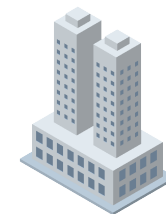
05 Make two sizes of recess



Make three columns by eight rows on each tower. On the mall, make seven columns by two rows of larger windows, then add the sides.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

M3 High/low housing + corner shops

Mixed use

Use the reference shape; dimensions are Width × Depth × Height (mm).

香港混合建築 (住宅+商舖)

建築外觀示意

Tinkercad 建模示意



AI 生成概念圖

Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | Two different-height towers share a corner-shop base.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base plate	86 × 64 × 2	×1
Corner shop base	74 × 52 × 18	×1
High residential tower	22 × 24 × 72	×1
Low residential tower	22 × 24 × 48	×1
Roof plant room	12 × 12 × 5	×2
Window-hole module (reused)	3 × 1 × 3	×1
Shopfront-hole module (reused)	6 × 1 × 11	×1

Challenge | Add more windows, a small entrance or a roof detail if time allows.

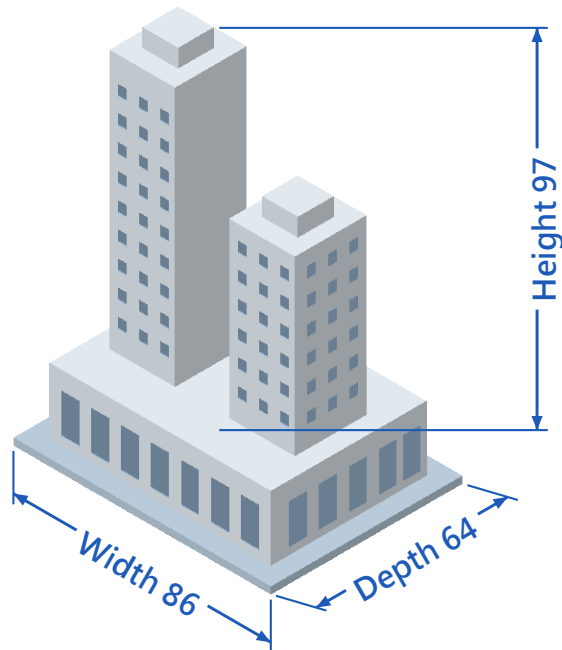
Print note | Check walls, bridges and roof parts with the teacher before printing.

M3 Three views and dimensions

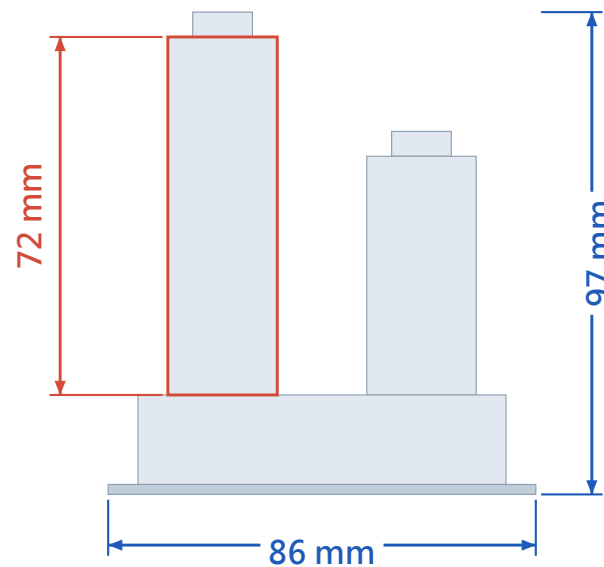
Dimension reference

Blue: overall (including base) | Red: High residential tower | Unit: mm; see the previous page for other parts.

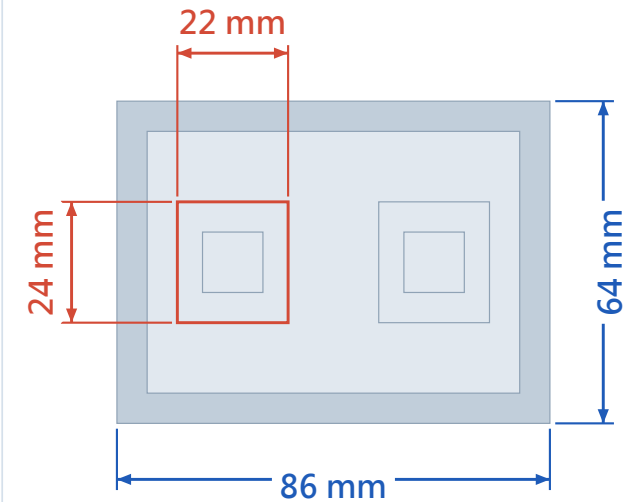
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = $86 \times 64 \times 97$ mm

High residential tower

Width × Depth × Height = $22 \times 24 \times 72$ mm

Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

M3 Build it: High/low housing + corner shops

Mixed use

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

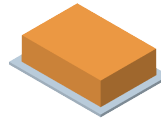
01 Make the base plate



Add a Box for the main part. Set the size shown in the parts list ($86 \times 64 \times 2$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

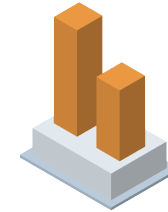
02 Place the shop base



Add the next Box using its listed size ($74 \times 52 \times 18$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

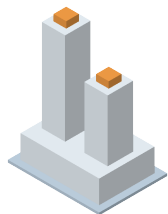
03 Place the high and low towers



Add or duplicate the next part ($22 \times 24 \times 72$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

04 Add each roof plant room



Add the roof, connector or entrance part ($22 \times 24 \times 48$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

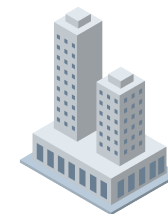
05 Make housing windows and two shop fronts



Make three columns on both towers: nine rows on the high tower and six on the low tower. Put seven shopfronts on the front and five on the side.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Group and check



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

C1 Modern office tower

Use the reference shape; dimensions are Width × Depth × Height (mm).

Commercial / high-rise

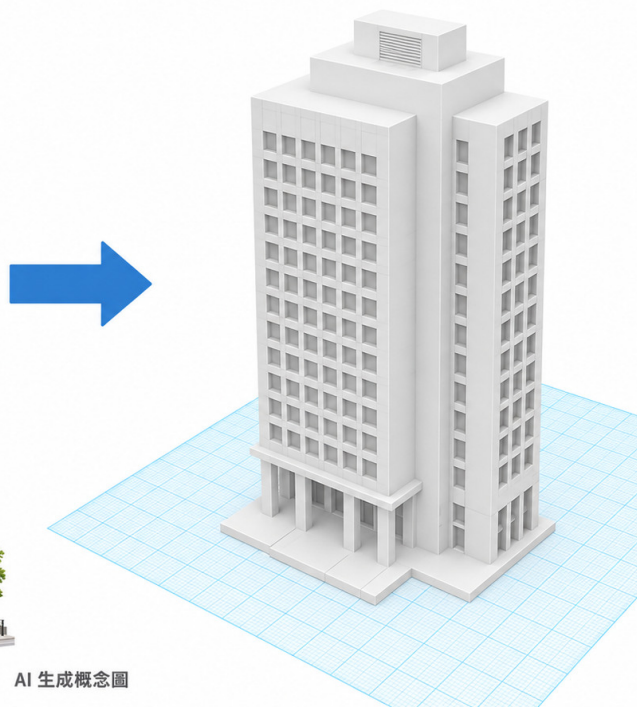
香港商業大廈 (現代辦公樓)

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | A tall and a shorter tower with a roof plant room and regular window rows.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base	60 × 42 × 3	×1
Main tower	26 × 26 × 84	×1
Side tower	14 × 22 × 72	×1
Roof slab	24 × 24 × 5	×1
Plant room	12 × 12 × 8	×1
Entrance canopy	24 × 8 × 2	×1
Recessed-window tool	3 × 1 × 4	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

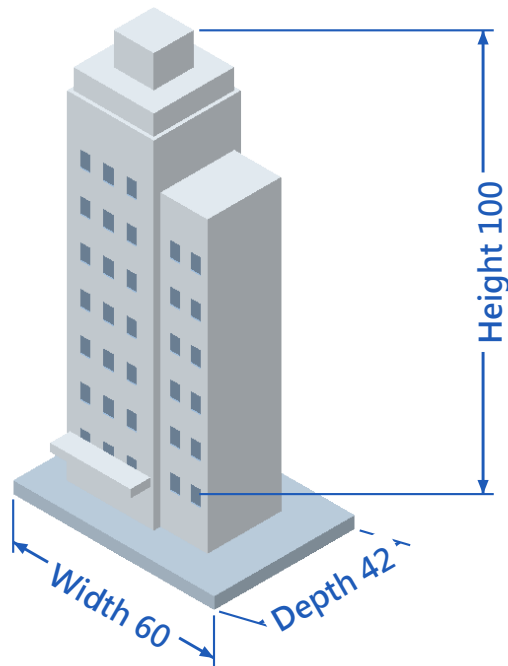
Print note | Check walls, bridges and roof parts with the teacher before printing.

C1 Three views and dimensions

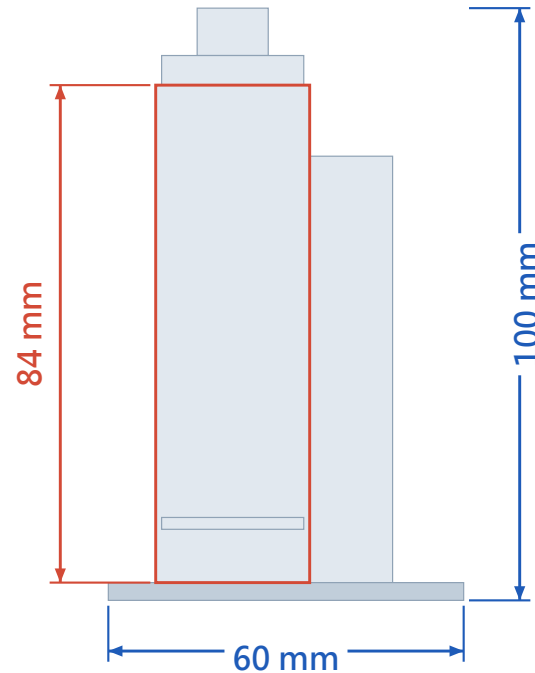
Dimension reference

Blue: overall (including base) | Red: Main tower | Unit: mm; see the previous page for other parts.

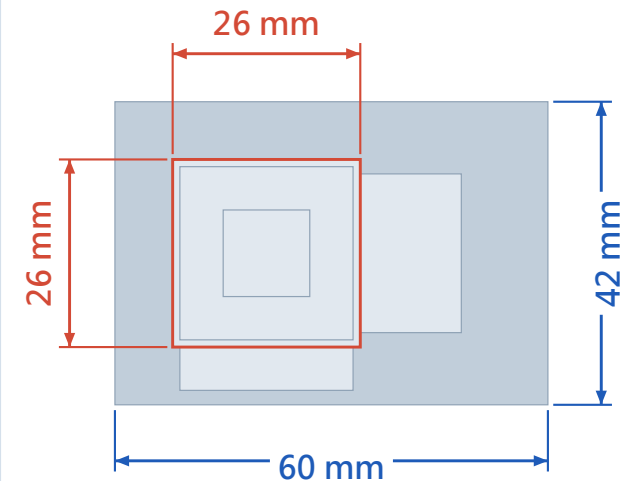
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = $60 \times 42 \times 100$ mm

Main tower

Width × Depth × Height = $26 \times 26 \times 84$ mm

Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

C1 Build it: Modern office tower

Commercial / high-rise

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

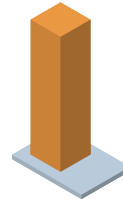
01 Place the base



Add a Box for the main part. Set the size shown in the parts list ($60 \times 42 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

02 Build the main tower



Add the next Box using its listed size ($26 \times 26 \times 84$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

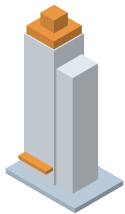
03 Add the side tower



Add or duplicate the next part ($14 \times 22 \times 72$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

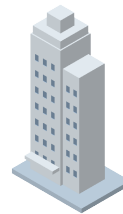
04 Add the roof and entrance



Add the roof, connector or entrance part ($24 \times 24 \times 5$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

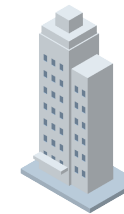
05 Duplicate recessed windows



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Check and group



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

C2 Mall-style commercial building

Use the reference shape; dimensions are Width × Depth × Height (mm).

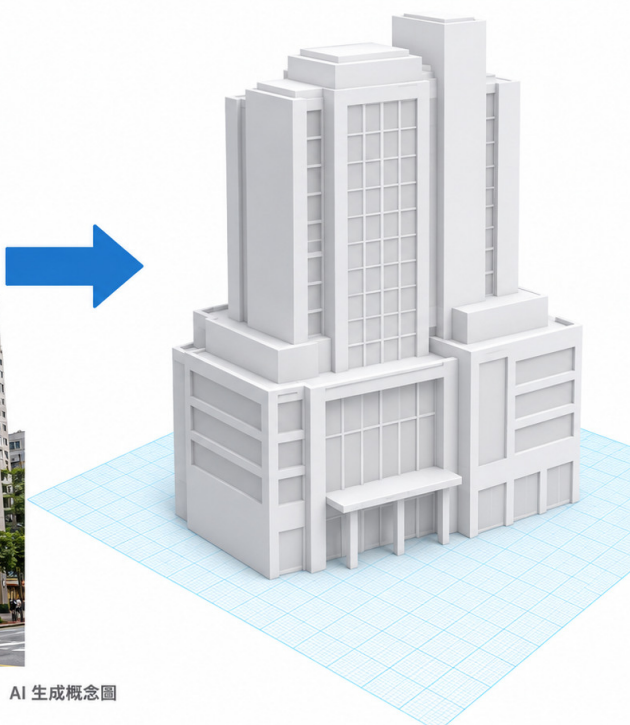
香港商業大廈（商場式商業樓）

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | A wide mall below a layered office tower.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Commercial / high-rise

Build dimensions

Base	78 × 56 × 3	×1
Mall	70 × 48 × 20	×1
Main tower	30 × 30 × 58	×1
Side wings	8 × 30 × 52 / 64	one e ach
Roof slab	24 × 24 × 6	×1
Entrance canopy	20 × 8 × 2	×1
Upper-window tool	4 × 1 × 5	reuse
Mall display-window tool	6 × 1 × 10	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

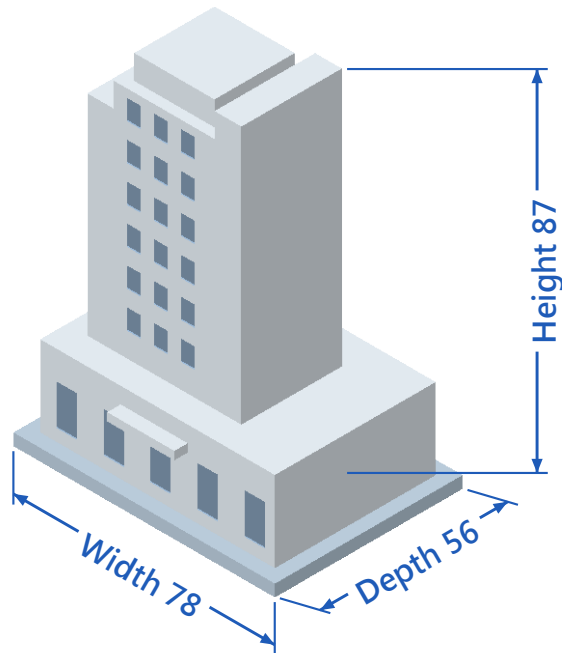
Print note | Check walls, bridges and roof parts with the teacher before printing.

C2 Three views and dimensions

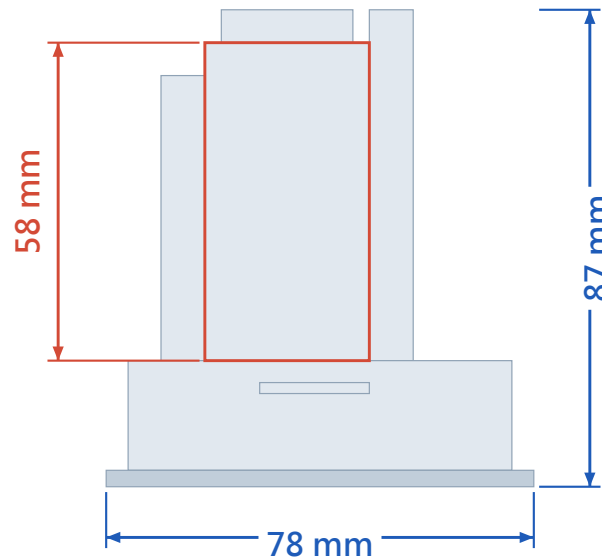
Dimension reference

Blue: overall (including base) | Red: Main tower | Unit: mm; see the previous page for other parts.

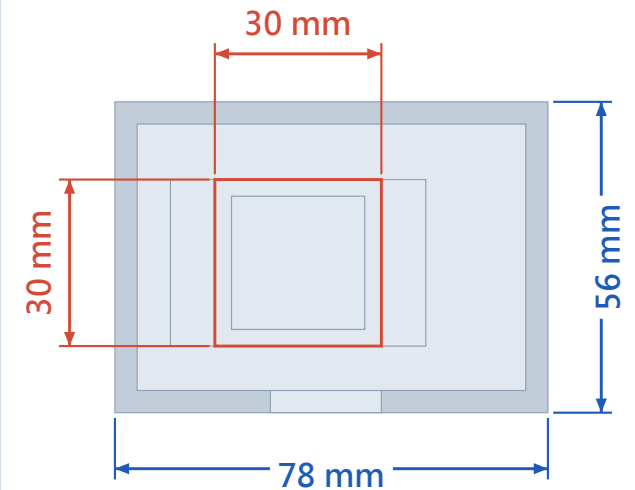
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 78 × 56 × 87 mm

Main tower

Width × Depth × Height = 30 × 30 × 58 mm

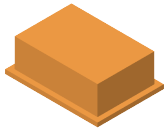
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

C2 Build it: Mall-style commercial building

Commercial / high-rise

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

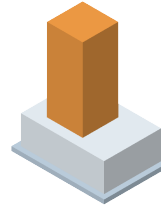
01 Build the mall



Add a Box for the main part. Set the size shown in the parts list ($78 \times 56 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

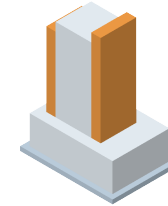
02 Place the main tower



Add the next Box using its listed size ($70 \times 48 \times 20$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

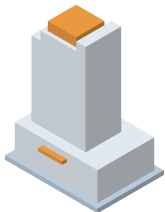
03 Add high and low wings



Add or duplicate the next part ($30 \times 30 \times 58$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

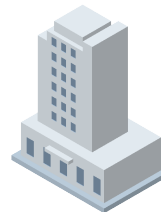
04 Add the roof and canopy



Add the roof, connector or entrance part ($8 \times 30 \times 52 / 64$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

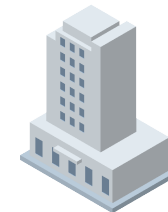
05 Make the two window types



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Check and group



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

C3 Slender pointed skyscraper

Use the reference shape; dimensions are Width × Depth × Height (mm).

Commercial / high-rise

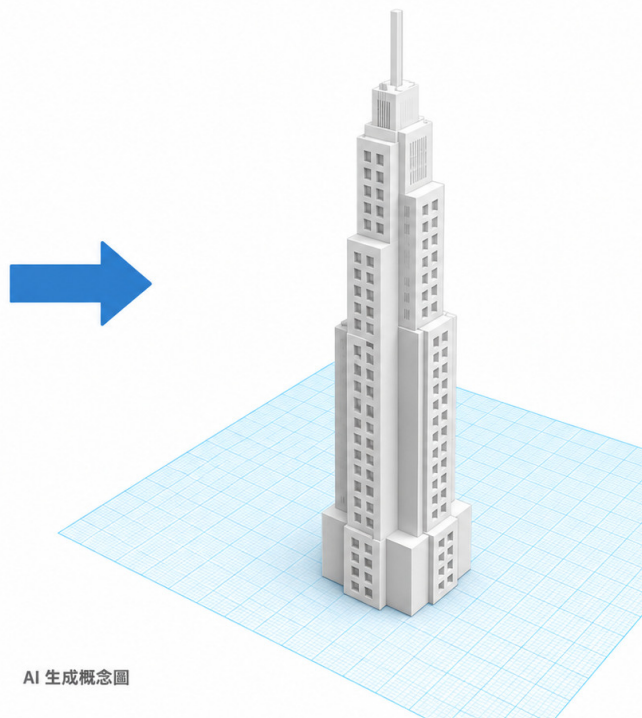
香港摩天大廈（纖細尖頂式）

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | The tower narrows upwards in steps and ends in a strong short spire.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base	44 × 44 × 3	×1
Lower tower	30 × 30 × 24	×1
Middle tower	24 × 24 × 36	×1
Upper tower	18 × 18 × 36	×1
Crown	12 × 12 × 12	×1
Spire	3 × 3 × 18	×1
Recessed-window tool	3 × 1 × 4	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

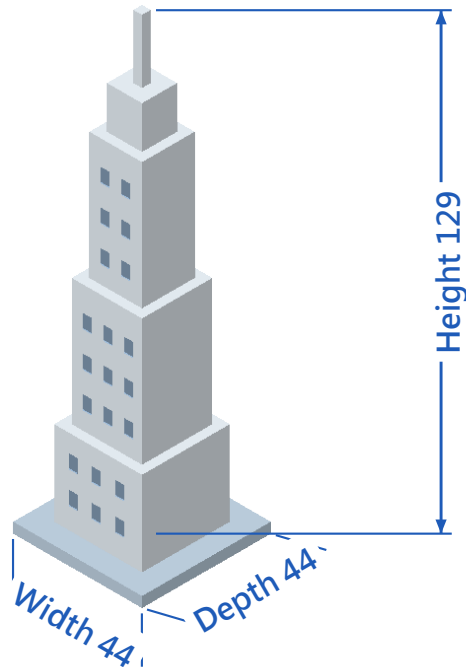
Print note | Check walls, bridges and roof parts with the teacher before printing.

C3 Three views and dimensions

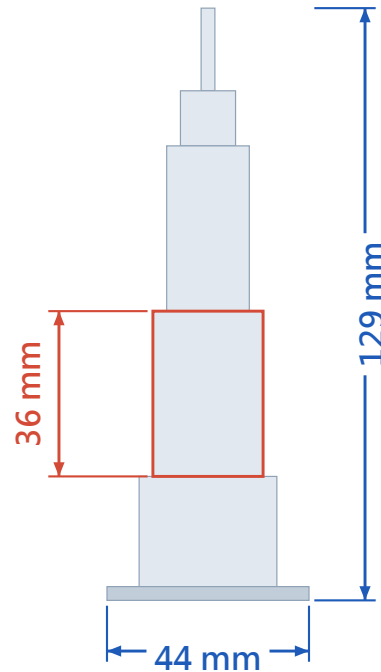
Dimension reference

Blue: overall (including base) | Red: Middle tower | Unit: mm; see the previous page for other parts.

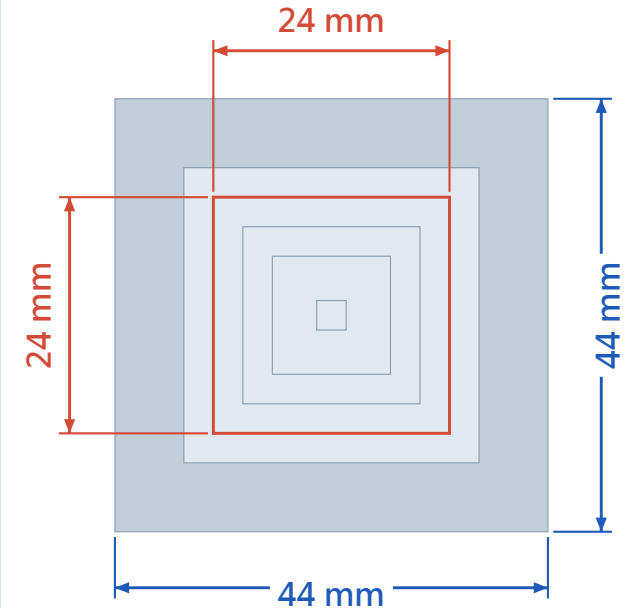
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 44 × 44 × 129 mm

Middle tower

Width × Depth × Height = 24 × 24 × 36 mm

Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

C3 Build it: Slender pointed skyscraper

Commercial / high-rise

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

01 Make the wide lower level



Add a Box for the main part. Set the size shown in the parts list ($44 \times 44 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

02 Narrow the tower layer by layer



Add the next Box using its listed size ($30 \times 30 \times 24$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

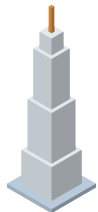
03 Add the tower crown



Add or duplicate the next part ($24 \times 24 \times 36$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

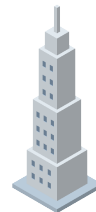
04 Add the strong short spire



Add the roof, connector or entrance part ($18 \times 18 \times 36$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

05 Make stepped window rows



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Check and group



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

C4 Twin glass towers

Use the reference shape; dimensions are Width × Depth × Height (mm).

Commercial / high-rise

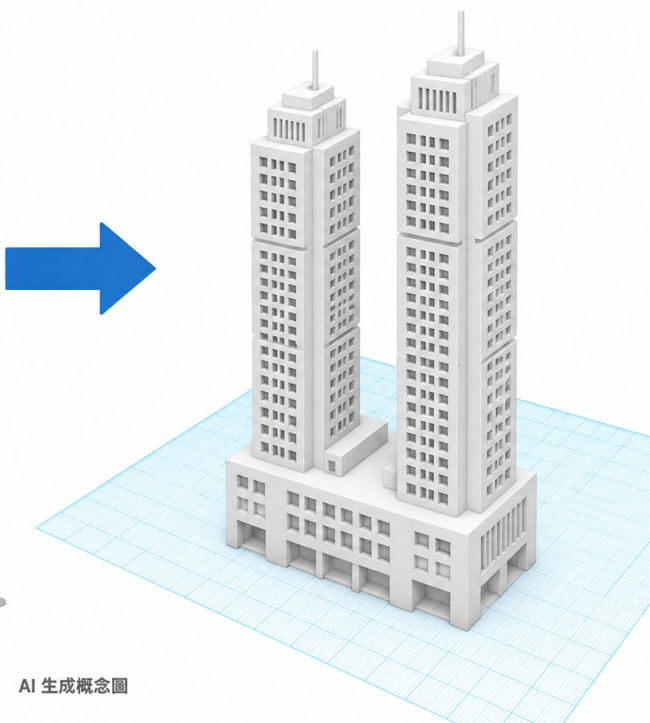
香港摩天大廈（雙塔玻璃式）

建築外觀示意



AI 生成概念圖

Tinkercad 建模示意



Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | Two different-height towers share a wide base, plant rooms and spires.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base	80 × 52 × 3	×1
Base building	72 × 44 × 12	×1
Left tower	20 × 20 × 72	×1
Right tower	20 × 20 × 90	×1
Roof slab	16 × 16 × 5	×2
Plant room	10 × 10 × 6	×2
Spire	3 × 3 × 12	×2
Recessed-window tool	3 × 1 × 4	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

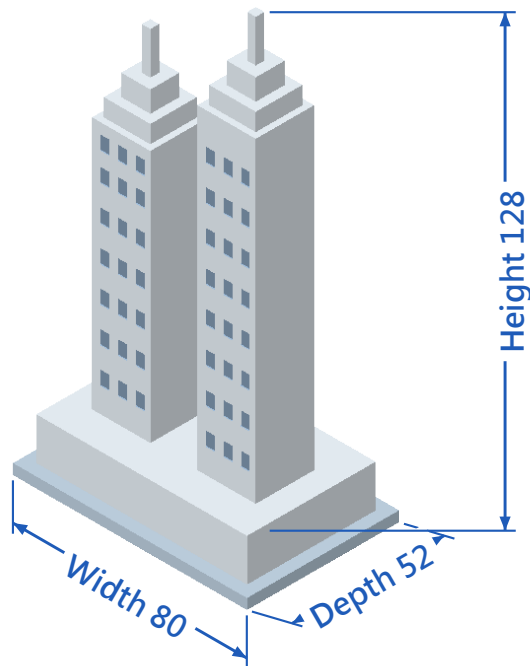
Print note | Check walls, bridges and roof parts with the teacher before printing.

C4 Three views and dimensions

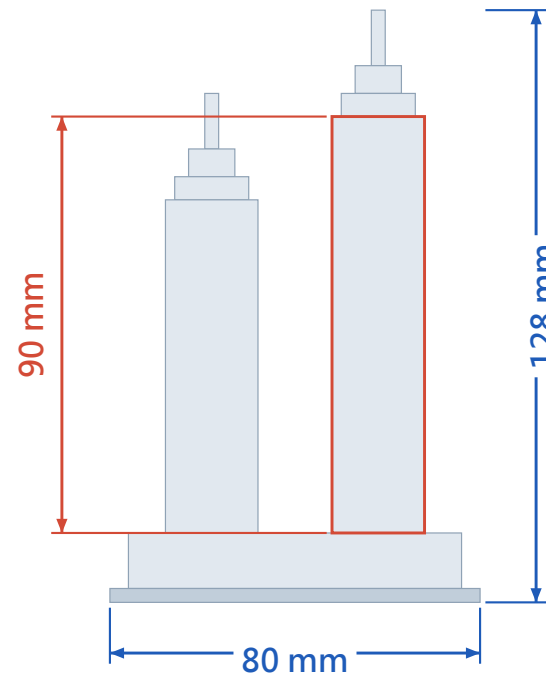
Dimension reference

Blue: overall (including base) | Red: Right tower | Unit: mm; see the previous page for other parts.

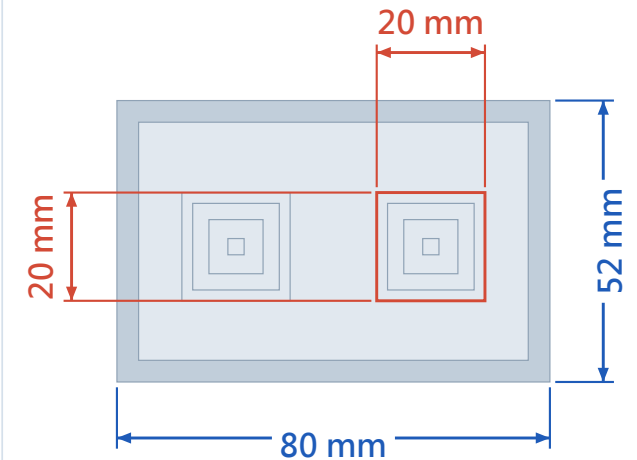
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = $80 \times 52 \times 128$ mm

Right tower

Width × Depth × Height = $20 \times 20 \times 90$ mm

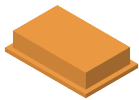
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

C4 Build it: Twin glass towers

Commercial / high-rise

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

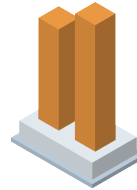
01 Make the shared base



Add a Box for the main part. Set the size shown in the parts list ($80 \times 52 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

02 Duplicate the two towers



Add the next Box using its listed size ($72 \times 44 \times 12$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

03 Add the two roof slabs



Add or duplicate the next part ($20 \times 20 \times 72$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

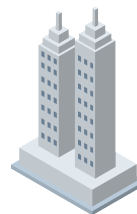
04 Add plant rooms and spires



Add the roof, connector or entrance part ($20 \times 20 \times 90$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

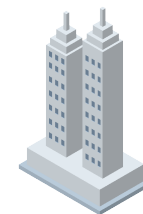
05 Add window rows



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Check and group



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

S1 Secondary school main block

Use the reference shape; dimensions are Width × Depth × Height (mm).

School

香港學校（中學主樓）

建築外觀示意



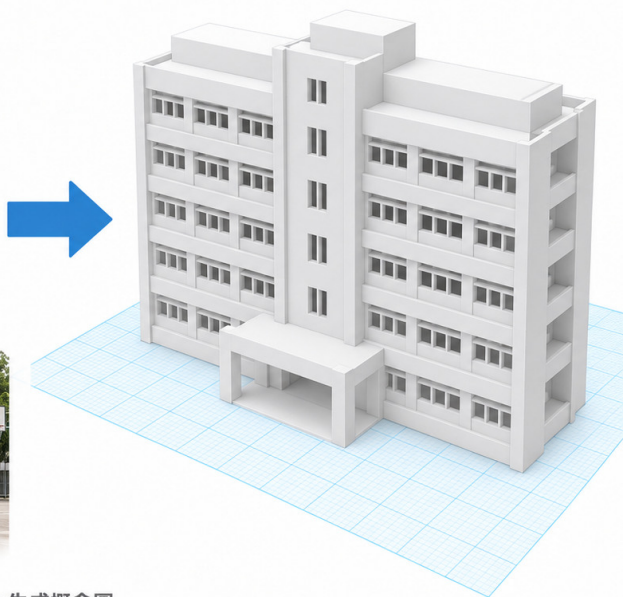
AI 生成概念圖

Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | A rectangular teaching block with a central stair tower and entrance canopy.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Tinkercad 建模示意



Build dimensions

Base	84 × 44 × 3	×1
Teaching block	72 × 20 × 40	×1
Stair tower	10 × 24 × 47	×1
Roof	74 × 22 × 2	×1
Entrance canopy	24 × 12 × 2	×1
Door columns	3 × 3 × 8	×2
Classroom-window tool	6 × 1 × 3	reuse
Stair-tower window tool	3 × 1 × 4	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

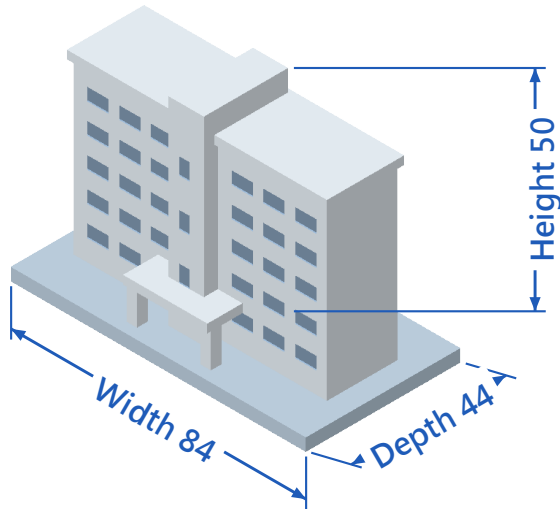
Print note | Check walls, bridges and roof parts with the teacher before printing.

S1 Three views and dimensions

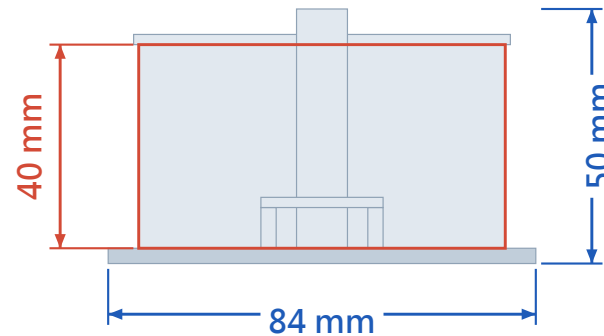
Dimension reference

Blue: overall (including base) | Red: Teaching block | Unit: mm; see the previous page for other parts.

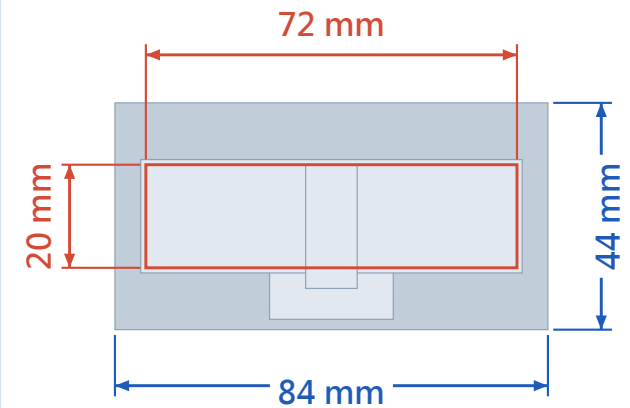
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 84 × 44 × 50 mm

Teaching block

Width × Depth × Height = 72 × 20 × 40 mm

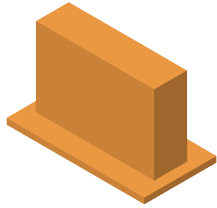
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

S1 Build it: Secondary school main block

School

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

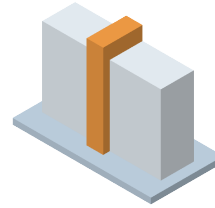
01 Make the teaching block



Add a Box for the main part. Set the size shown in the parts list ($84 \times 44 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

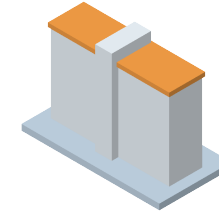
02 Add the stair tower



Add the next Box using its listed size ($72 \times 20 \times 40$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

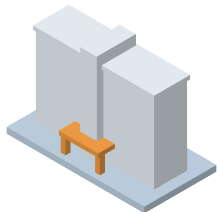
03 Place the roof



Add or duplicate the next part ($10 \times 24 \times 47$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

04 Make the columned entrance



Add the roof, connector or entrance part ($74 \times 22 \times 2$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

05 Duplicate classroom windows



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Check and group



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

S2 L-shaped school campus

Use the reference shape; dimensions are Width × Depth × Height (mm).

School

香港學校（L型校舍）

建築外觀示意

Tinkercad 建模示意



AI 生成概念圖

Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | Two wings meet at a right angle around an open playground.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base	80 × 80 × 3	×1
Rear wing	64 × 16 × 36	×1
Left wing	16 × 64 × 36	×1
Stair tower	18 × 18 × 44	×1
Rear-wing roof	66 × 18 × 2	×1
Left-wing roof	18 × 66 × 2	×1
Classroom-window tool	5 × 1 × 4	reuse
Stair-tower window tool	3 × 1 × 4	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

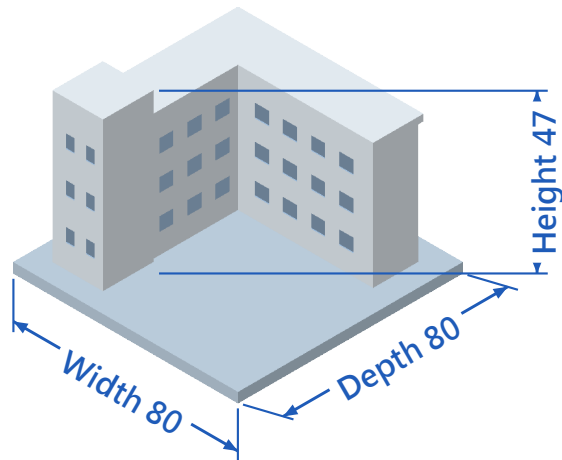
Print note | Check walls, bridges and roof parts with the teacher before printing.

S2 Three views and dimensions

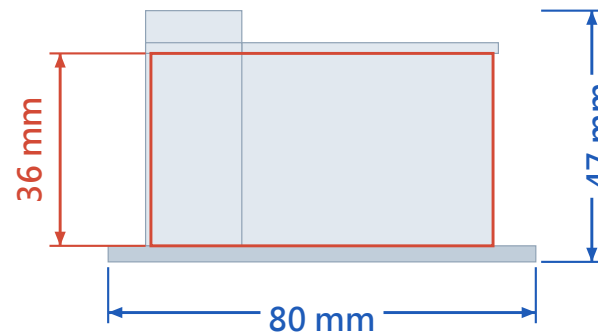
Dimension reference

Blue: overall (including base) | Red: Rear wing | Unit: mm; see the previous page for other parts.

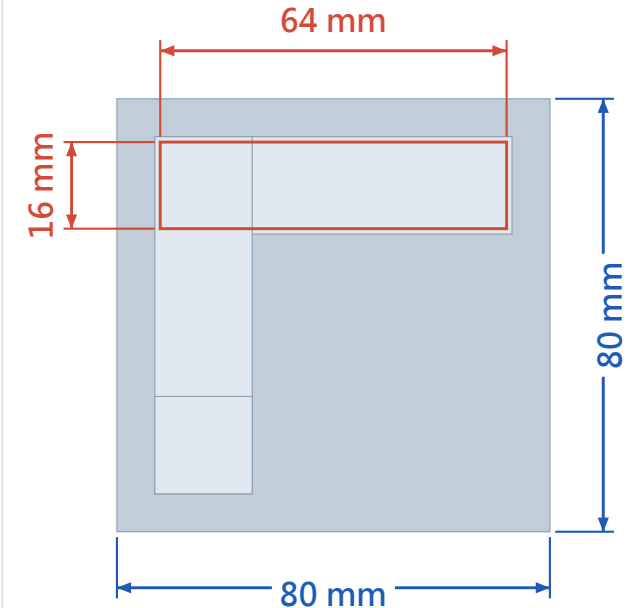
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 80 × 80 × 47 mm

Rear wing

Width × Depth × Height = 64 × 16 × 36 mm

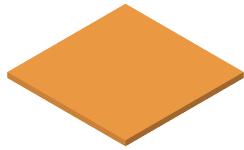
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

S2 Build it: L-shaped school campus

School

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

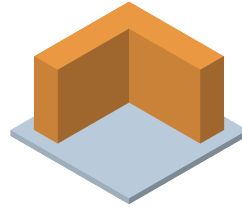
01 Leave the playground open



Add a Box for the main part. Set the size shown in the parts list ($80 \times 80 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

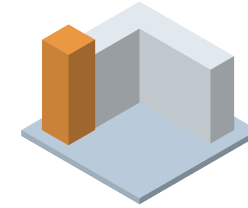
02 Arrange an L shape



Add the next Box using its listed size ($64 \times 16 \times 36$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

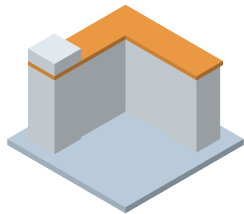
03 Add the stair tower



Add or duplicate the next part ($16 \times 64 \times 36$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

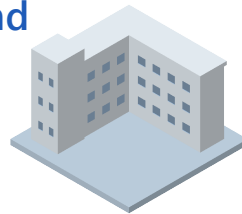
04 Add both wing roofs



Add the roof, connector or entrance part ($18 \times 18 \times 44$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

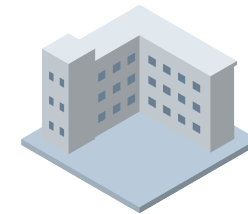
05 Open windows towards the playground



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Check and group



Select the solid parts with Shift, press Ctrl+G (Group / Union), then check the model from TOP and FRONT. Change the residential version to blue and the other versions to white.

Check: The building is connected, with no floating pieces or unselected parts.

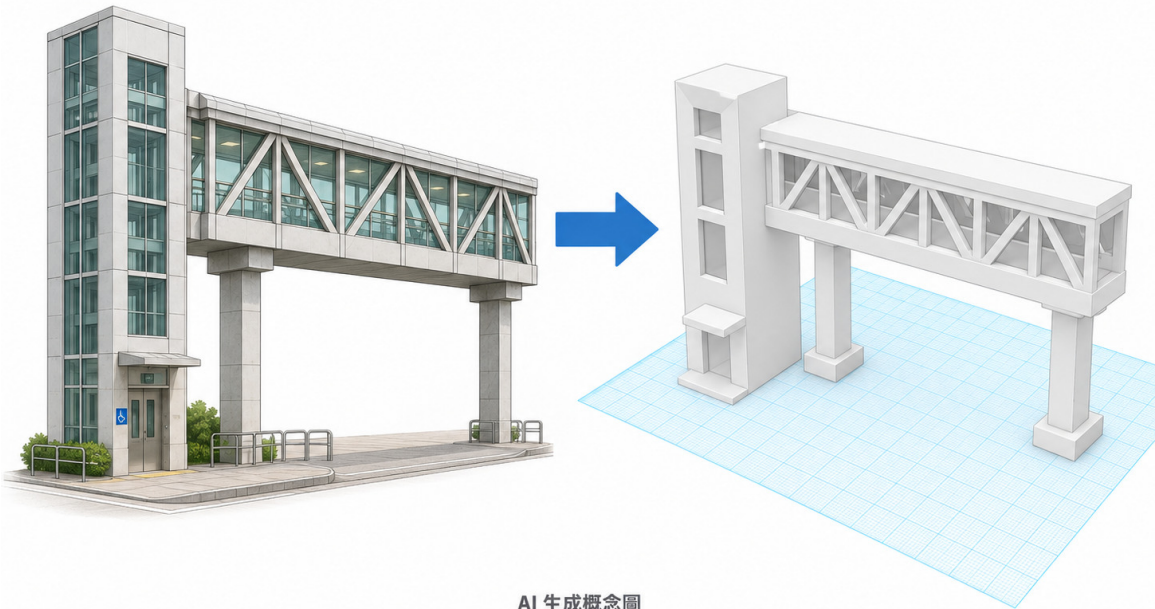
B1 Straight covered footbridge

Use the reference shape; dimensions are Width × Depth × Height (mm).

香港有蓋行人天橋

建築外觀示意

Tinkercad 建模示意



AI 生成概念圖

Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | Thick side walls, square columns and a flat roof connect to one lift tower.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Footbridge

Build dimensions

Base	128 × 42 × 3	×1
Lift tower	16 × 20 × 42	×1
Bridge columns	6 × 8 × 24	×2
Bridge deck	108 × 16 × 3	×1
Side walls	108 × 2 × 12	×2
Top cover	110 × 20 × 3	×1
Bridge-window tool	8 × 1 × 6	reuse
Tower-window tool	6 × 1 × 7	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

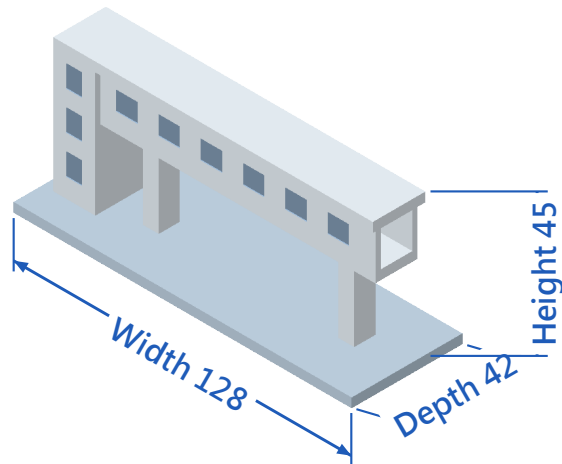
Print note | Check walls, bridges and roof parts with the teacher before printing.

B1 Three views and dimensions

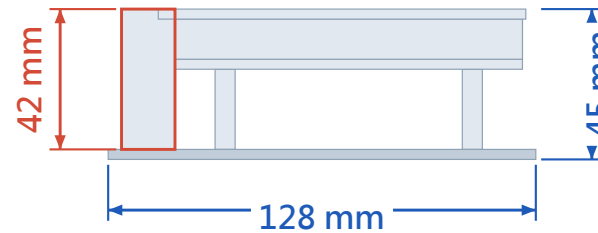
Dimension reference

Blue: overall (including base) | Red: Lift tower | Unit: mm; see the previous page for other parts.

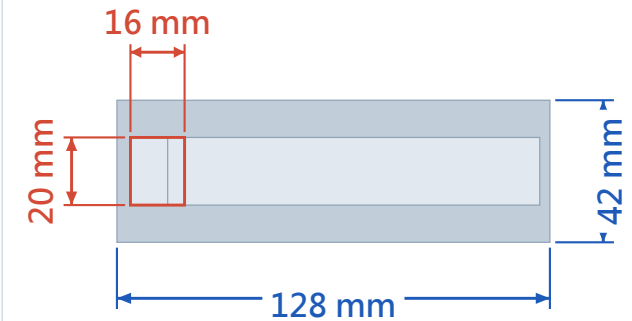
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 128 × 42 × 45 mm

Lift tower

Width × Depth × Height = 16 × 20 × 42 mm

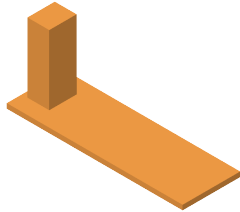
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

B1 Build it: Straight covered footbridge

Footbridge

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

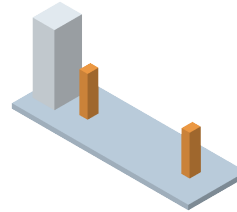
01 Make the base and tower



Add a Box for the main part. Set the size shown in the parts list ($128 \times 42 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

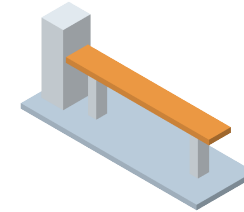
02 Place the two bridge columns



Add the next Box using its listed size ($16 \times 20 \times 42$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

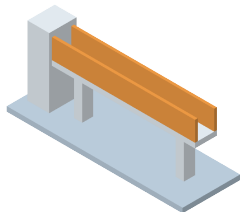
03 Add the bridge deck



Add or duplicate the next part ($6 \times 8 \times 24$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

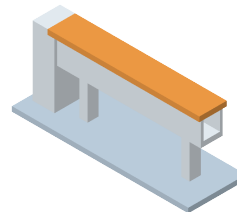
04 Add the two side walls



Add the roof, connector or entrance part ($108 \times 16 \times 3$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

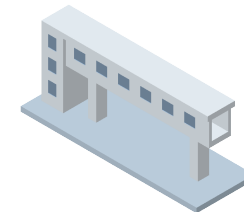
05 Add the top cover



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Make windows and finish



Add a few shallow recesses to the bridge and tower. Ask the teacher to keep a separate, ungrouped version before making the final Union.

Check: The building is connected, with no floating pieces or unselected parts.

B2 Covered footbridge with stairs

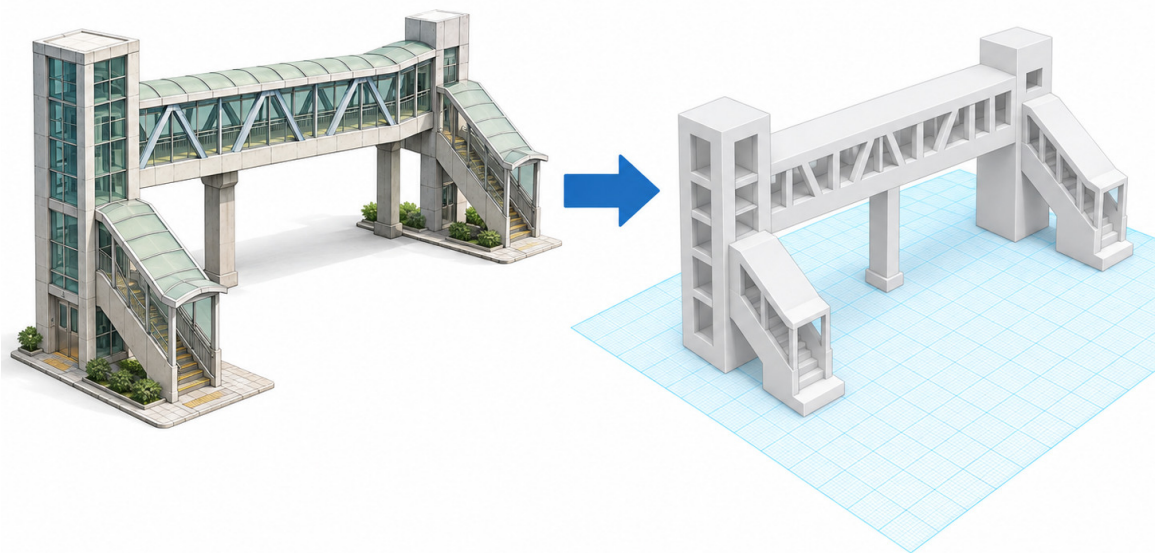
Use the reference shape; dimensions are Width × Depth × Height (mm).

Footbridge

香港有蓋行人天橋

建築外觀示意

Tinkercad 建模示意



AI 生成概念圖

Teacher-provided AI concept reference; see the next page for the three views and arrow dimensions.

Observation focus | Two lift towers and a covered bridge with six stairs at each end.

Basic version: complete the main form first. Add a few windows, then add more if time allows.

Build dimensions

Base	128 × 70 × 3	×1
Lift tower	12 × 18 × 46	×2
Central bridge column	6 × 8 × 30	×1
Bridge deck	112 × 14 × 3	×1
Side walls	112 × 2 × 10	×2
Top cover	114 × 18 × 3	×1
Six-step stair set at each end	W 12, D 6; H 5.5, add 5.5	2 sets
Bridge-window tool	8 × 1 × 6	reuse

Challenge | Add more windows, a small entrance or a roof detail if time allows.

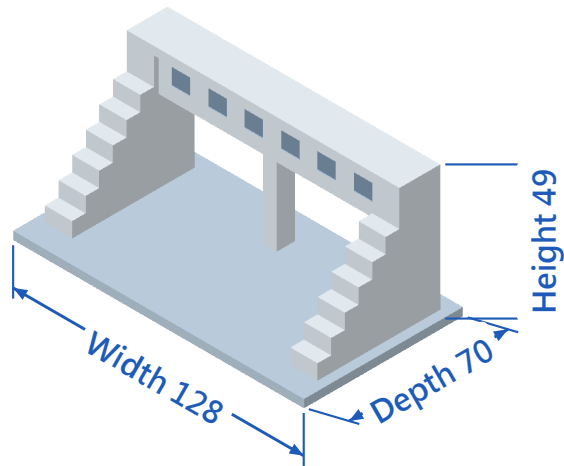
Print note | Check walls, bridges and roof parts with the teacher before printing.

B2 Three views and dimensions

Dimension reference

Blue: overall (including base) | Red: Left tower | Unit: mm; see the previous page for other parts.

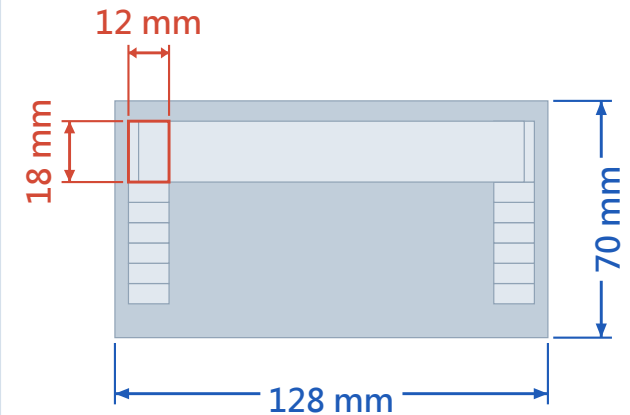
1. Isometric view



2. Front view



3. Top view



Overall (including base)

Width × Depth × Height = 128 × 70 × 49 mm

Left tower

Width × Depth × Height = 12 × 18 × 46 mm

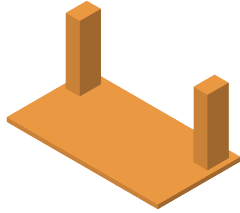
Arrows show model dimensions, not placement coordinates. Views are scaled to fit; use the labelled numbers.

B2 Build it: Covered footbridge with stairs

Footbridge

Orange = new solid in this step; dark squares = recessed-window examples. Cell 6 shows the finished outline.

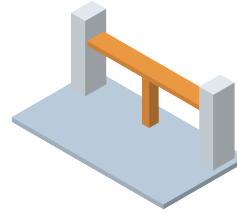
01 Make the two end towers



Add a Box for the main part. Set the size shown in the parts list ($128 \times 70 \times 3$ mm), then place it on the base. Use Workplane and D so it touches the surface.

Check: The part is thin, stable and inside the base.

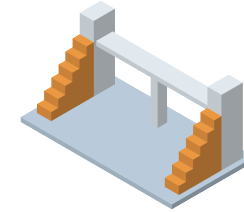
02 Add the covered bridge



Add the next Box using its listed size ($12 \times 18 \times 46$ mm). Place it on the previous part and keep the building inside the base.

Check: The main block touches the base and leaves a clear outline.

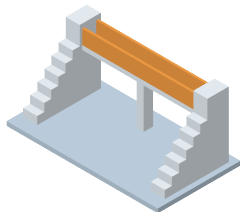
03 Build the stairs step by step



Add or duplicate the next part ($6 \times 8 \times 30$ mm). Arrange the copies as shown in the reference and let neighbouring faces overlap slightly.

Check: The pieces are aligned, slightly overlapped and at the same support height.

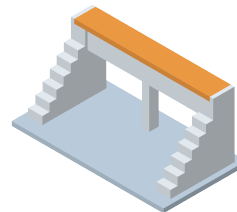
04 Make the bridge side walls



Add the roof, connector or entrance part ($112 \times 14 \times 3$ mm). Use Workplane on the supporting top face, then press D so the new part touches it.

Check: The new roof, bridge or entrance touches the main body.

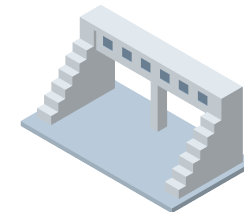
05 Place the flat top cover



Make a small Hole, duplicate it into the window pattern and keep every recess inside the wall. Leave at least 2 mm of solid wall.

Check: The windows stay inside the walls and the main silhouette remains readable.

06 Add windows and finish



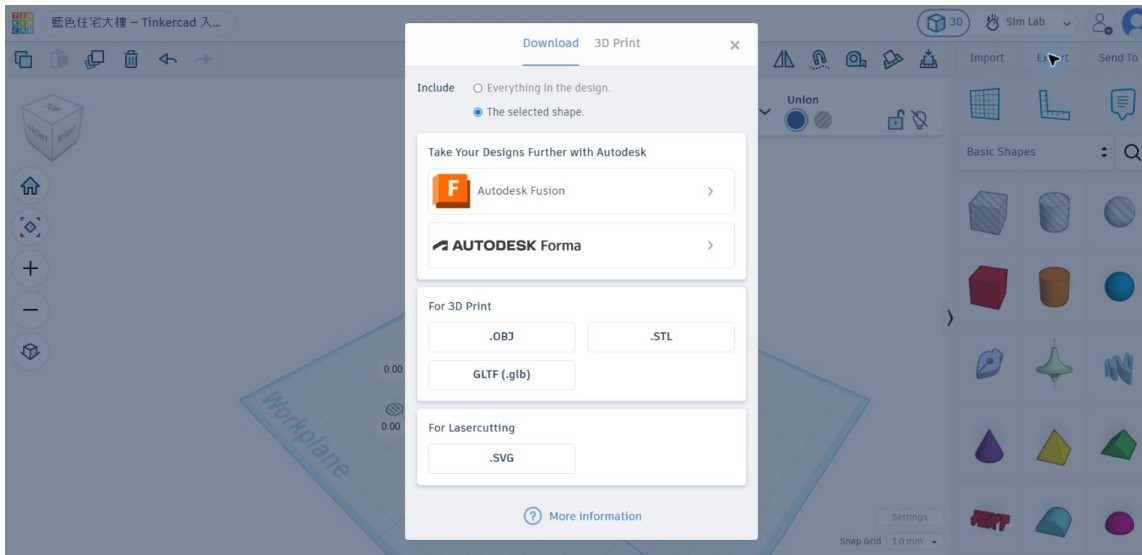
Add a few shallow recesses to the bridge and tower. Ask the teacher to keep a separate, ungrouped version before making the final Union.

Check: The building is connected, with no floating pieces or unselected parts.

Export STL: give the model to the teacher

[Submission reference](#)

Keep this export reference and use it after your model is complete.



1 | Click Export

Select the complete model and click Export at the top right. If you are submitting only the selection, choose The selected shape.

2 | Download .STL

Under For 3D Print, click .STL.

Filename: Class_Group_ModelID_Quantity

Example: 2B_GROUP 2_B1_1.stl

(2B class, Group 2, model B1, one piece)

3 | Submit a completion screenshot

The screenshot must show the complete form. The teacher imports the STL in millimetres and checks thickness, supports and print direction.